Advanced Reinforcement learning and application in robotics and LLM

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[Tailin Wu](), Westlake University

Website: [ai4s.lab.westlake.edu.cn/course]()

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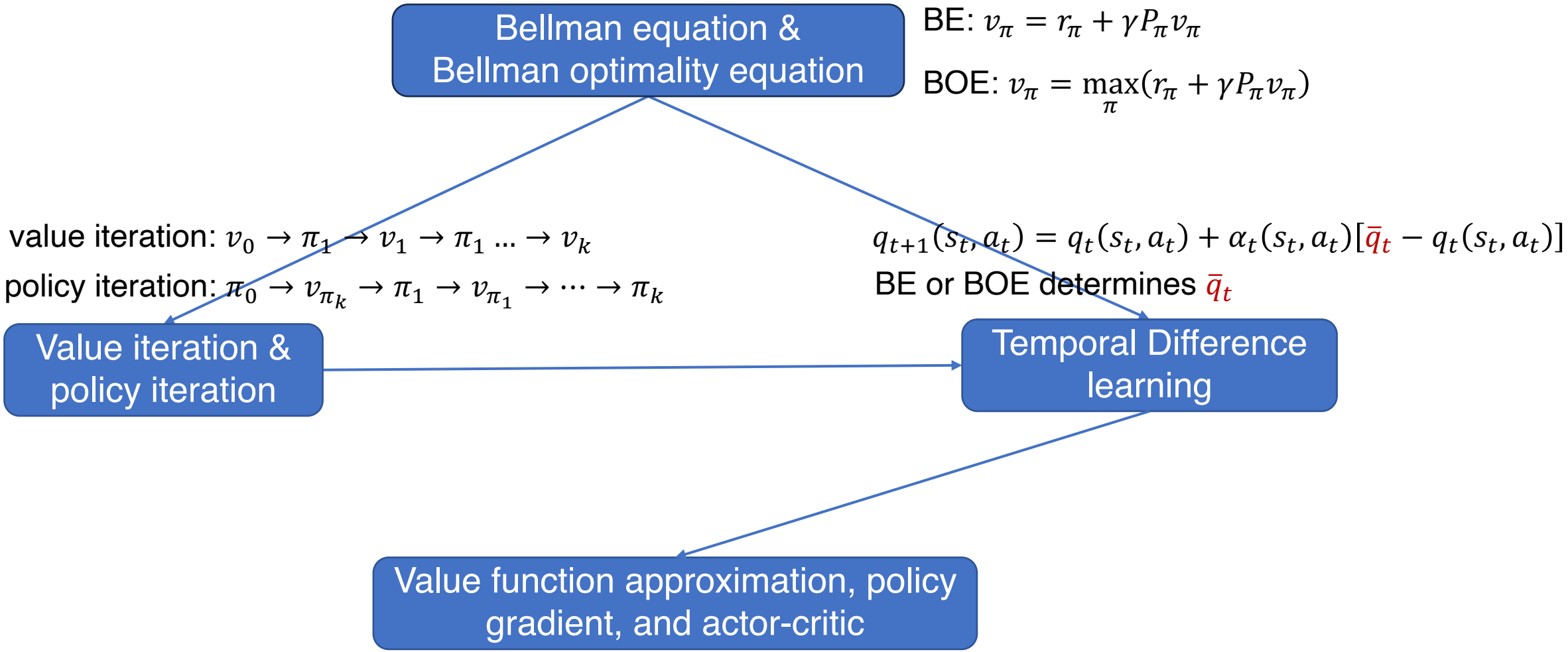
Outline

- ❑ Deep Reinforcement Learning
- ❑ Applications to Robotics
- ❑ Applications to LLMs

Deep Reinforcement Learning (DRL)

- Model-free DRL
- Model-based DRL
- Inverse Reinforcement Learning
- Offline Reinforcement Learning
- Large Pre-training DRL Model



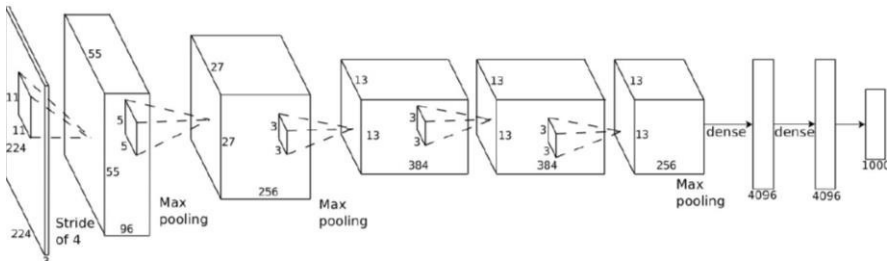
TD with function approximation: $w_{t+1} = w_t + \alpha_t\left[\left(r_{t+1} + \gamma\hat{v}(s_{t+1}; w_t)\right) - \hat{v}(s_t; w_t)\right]\nabla_w \hat{v}(s_t; w_t)$

REINFORCE: $\theta_{t+1} = \theta_t + \alpha\nabla_\theta \ln \pi(a_t|s_t; \theta_t) q_t(s_t, a_t)$

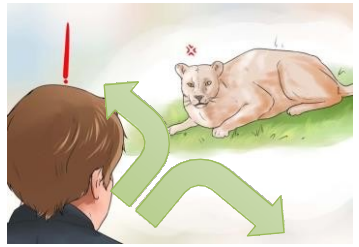
Actor-critic: $\theta_{t+1} = \theta_t + \alpha\nabla_\theta \ln \pi(a_t|s_t; \theta_t)\left(r_{t+1} + \gamma v(s_{t+1}; w_t) - v_t(s_t; w_t)\right)$

Terminology & Notation of DRL

$\mathbf{o}_t$



$\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$



$\mathbf{a}_t$

$\mathbf{s}_t$ – state
$\mathbf{o}_t$ – observation
$\mathbf{a}_t$ – action

$\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ – policy
$\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ – policy (fully observed)

State value

State value: Suppose that we start with a state s and follow the policy $\pi(a|s)$, state value is the **expected discounted return**

$$v_\pi(s) \coloneqq \mathbb{E}[G_t|S_t = s] = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots |S_t = s]$$

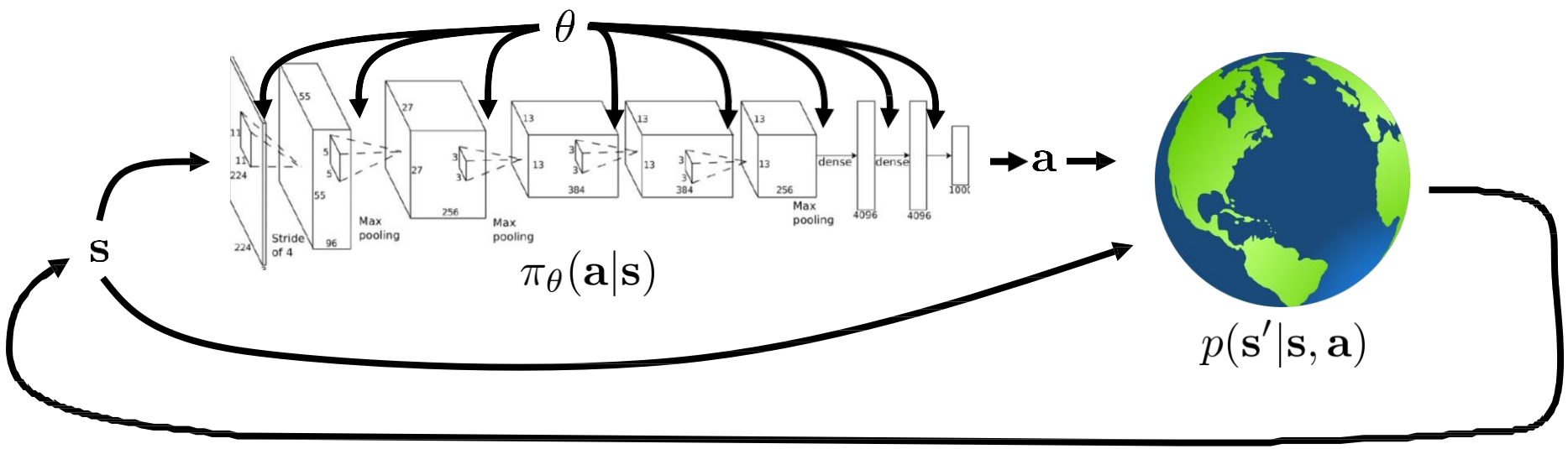
The state value function evaluates how good the policy π is.

Action value: Definition

$$q_\pi(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a]$$

The average return the agent can get starting from a state s and taking an action a, and then following the policy π.

Goal of DRL



$$\underbrace{p_\theta(\mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_T, \mathbf{a}_T)}_{p_\theta(\tau)} = p(\mathbf{s}_1) \prod_{t=1}^{T} \pi_\theta(\mathbf{a}_t|\mathbf{s}_t) p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$$

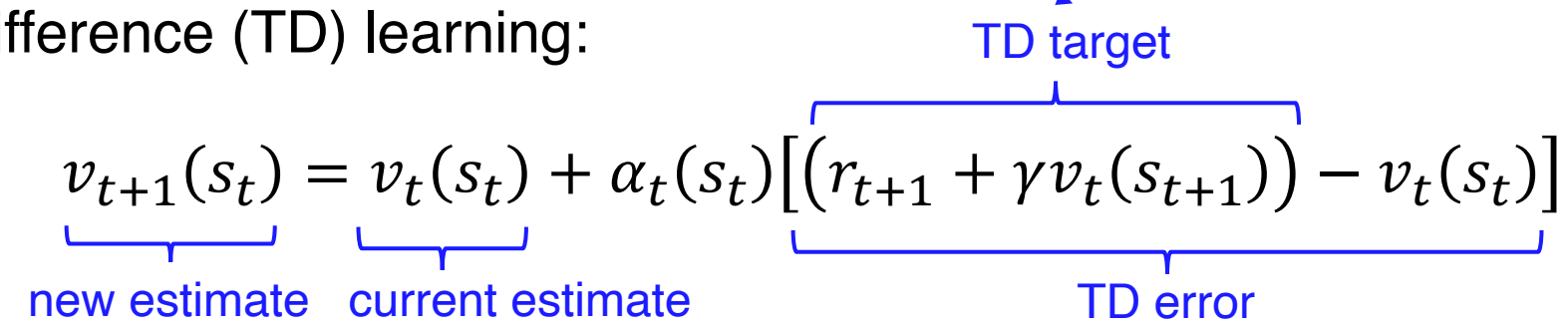
$$\theta^\star = \arg\max_\theta E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

Temporal difference learning

Bellman equation (expected form):

$$v_\pi(s) = \mathbb{E}[R + \gamma v_\pi(S') | S = s]$$

Temporal difference (TD) learning:



$$v_{t+1}(s_t) = v_t(s_t) + \alpha_t(s_t)\left[\left(r_{t+1} + \gamma v_t(s_{t+1})\right) - v_t(s_t)\right]$$

- It uses boostrapping, since $r_{t+1} + \gamma v_t(s_{t+1})$ is a better estimate of $v_\pi(s_t)$ than $v_t(s_t)$.

Other temporal difference learning: Unified view

We also have Bellman equation (BE) and Bellman optimality equation (BOE) for the action value $q(s,a)$, and have corresponding TD learning algorithm:

$$q_{t+1}(s_t,a_t) = q_t(s_t,a_t) + \alpha_t(s_t,a_t)[\bar{q}_t - q_t(s_t,a_t)]$$

Algorithm	Equation and expression of $\bar{q}_t$
Sarsa	BE: $q_\pi(s,a) = \mathbb{E}[R_{t+1} + \gamma q_\pi(S_{t+1},A_{t+1})\|S_t = s, A_t = a]$
	$\bar{q}_t = r_{t+1} + \gamma q(s_{t+1},a_{t+1})$
n-step Sarsa	BE: $q_\pi(s,a) = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^n q_\pi(S_{t+n},A_{t+n})\|S_t = s, A_t = a]$
	$\bar{q}_t = r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^n q(s_{t+n},a_{t+n})$
Q-learning	BOE: $q_\pi(s,a) = \mathbb{E}\left[R_{t+1} + \gamma \max_a q_\pi(S_{t+1},a)\,\middle\|\, S_t = s, A_t = a\right]$
	$\bar{q}_t = r_{t+1} + \gamma \max_a q(s_{t+1},a)$
Monte Carlo	BE: $q_\pi(s,a) = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \cdots \|S_t = s, A_t = a]$
	$\bar{q}_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} \ldots$

Policy Gradients

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t}|\mathbf{s}_{i,t}) \underbrace{\left(\sum_{t'=t}^{T} r(\mathbf{s}_{i,t'}, \mathbf{a}_{i,t'}) \right)}_{\text{“reward to go”} \ \hat{Q}_{i,t}}$$

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t}|\mathbf{s}_{i,t}) Q(\mathbf{s}_{i,t}, \mathbf{a}_{i,t})$$

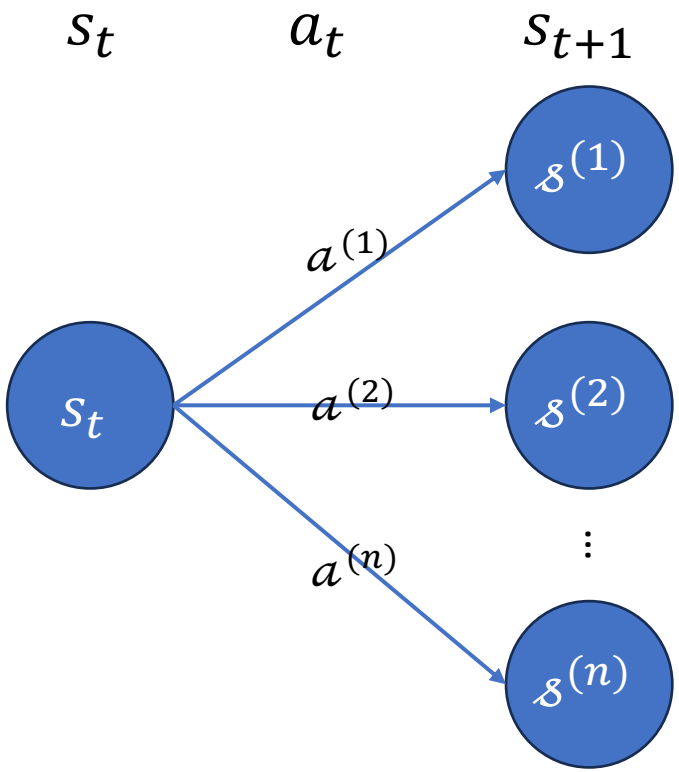
Baseline: $V(\mathbf{s}_t) = E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}[Q(\mathbf{s}_t, \mathbf{a}_t)]$

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t}|\mathbf{s}_{i,t}) \left(Q(\mathbf{s}_{i,t}, \mathbf{a}_{i,t}) - V(\mathbf{s}_{i,t})\right)$$

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t}|\mathbf{s}_{i,t}) A(\mathbf{s}_{i,t}, \mathbf{a}_{i,t})$$

Actor-critic: Interpretation

$$\theta_{t+1} = \theta_t + \alpha \nabla_\theta \ln \pi(a_t|s_t;\theta_t)\overbrace{\left(r_{t+1} + \gamma v(s_{t+1};w_t) - v_t(s_t;w_t)\right)}^{\text{Advantage } \delta_t}$$



δ_t

0.5 $\quad \theta_{t+1} = \theta_t + \alpha \nabla_\theta \ln \pi\left(a_t = a^{(1)}|s_t;\theta_t\right) \cdot 0.5$ ↑ increase likelihood

-0.2 $\quad \theta_{t+1} = \theta_t + \alpha \nabla_\theta \ln \pi\left(a_t = a^{(2)}|s_t;\theta_t\right) \cdot (-0.2)$ ↓ decrease likelihood

-1 $\quad \theta_{t+1} = \theta_t + \alpha \nabla_\theta \ln \pi\left(a_t = a^{(n)}|s_t;\theta_t\right) \cdot (-1)$ ↓ decrease likelihood

Value function Fitting

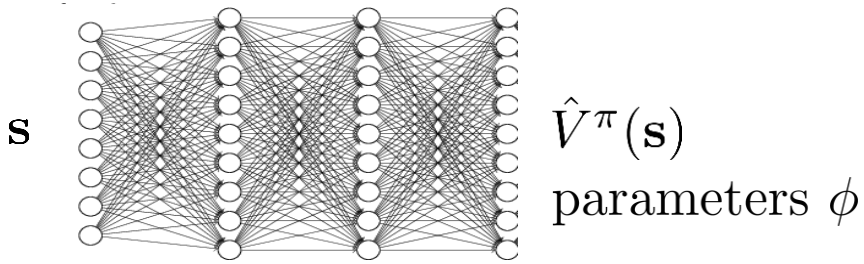
$$\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t}|\mathbf{s}_{i,t})A^\pi(\mathbf{s}_{i,t},\mathbf{a}_{i,t})$$

$$A^\pi(\mathbf{s}_t,\mathbf{a}_t) = Q^\pi(\mathbf{s}_t,\mathbf{a}_t) - V^\pi(\mathbf{s}_t)$$

fit *what* to *what*?

$$Q^\pi(\mathbf{s}_t,\mathbf{a}_t) \approx r(\mathbf{s}_t,\mathbf{a}_t) + V^\pi(\mathbf{s}_{t+1})$$

$$A^\pi(\mathbf{s}_t,\mathbf{a}_t) \approx r(\mathbf{s}_t,\mathbf{a}_t) + V^\pi(\mathbf{s}_{t+1}) - V^\pi(\mathbf{s}_t)$$



I: $y_{i,t} \approx \sum_{t'=t}^{T} r(\mathbf{s}_{i,t'},\mathbf{a}_{i,t'})$

II: $y_{i,t} \approx r(\mathbf{s}_{i,t},\mathbf{a}_{i,t}) + \hat{V}_\phi^\pi(\mathbf{s}_{i,t+1})$

$$\mathcal{L}(\phi) = \frac{1}{2}\sum_i \left\|\hat{V}_\phi^\pi(\mathbf{s}_i) - y_i\right\|^2$$

Actor-Critic Algorithm (with Discount)

batch actor-critic algorithm:

1. sample $\{\mathbf{s}_i, \mathbf{a}_i\}$ from $\pi_\theta(\mathbf{a}|\mathbf{s})$ (run it on the robot)
2. fit $\hat{V}^\pi_\phi(\mathbf{s})$ to sampled reward sums
3. evaluate $\hat{A}^\pi(\mathbf{s}_i, \mathbf{a}_i) = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \hat{V}^\pi_\phi(\mathbf{s}'_i) - \hat{V}^\pi_\phi(\mathbf{s}_i)$
4. $\nabla_\theta J(\theta) \approx \sum_i \nabla_\theta \log \pi_\theta(\mathbf{a}_i|\mathbf{s}_i) \hat{A}^\pi(\mathbf{s}_i, \mathbf{a}_i)$
5. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

online actor-critic algorithm:

1. take action $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$, get $(\mathbf{s}, \mathbf{a}, \mathbf{s}', r)$
2. update $\hat{V}^\pi_\phi$ using target $r + \gamma \hat{V}^\pi_\phi(\mathbf{s}')$
3. evaluate $\hat{A}^\pi(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \gamma \hat{V}^\pi_\phi(\mathbf{s}') - \hat{V}^\pi_\phi(\mathbf{s})$
4. $\nabla_\theta J(\theta) \approx \nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s}) \hat{A}^\pi(\mathbf{s}, \mathbf{a})$
5. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

Run policy:

1. Collect data
2. Fit value function

Update policy:

1. Evaluate Advantage A
2. Gradient Descent

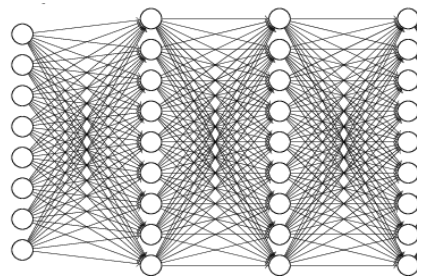
“Max” Trick to Remove Policy Gradient

Max Trick for Q-Value

1. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)\}$ using some policy

$K\times$

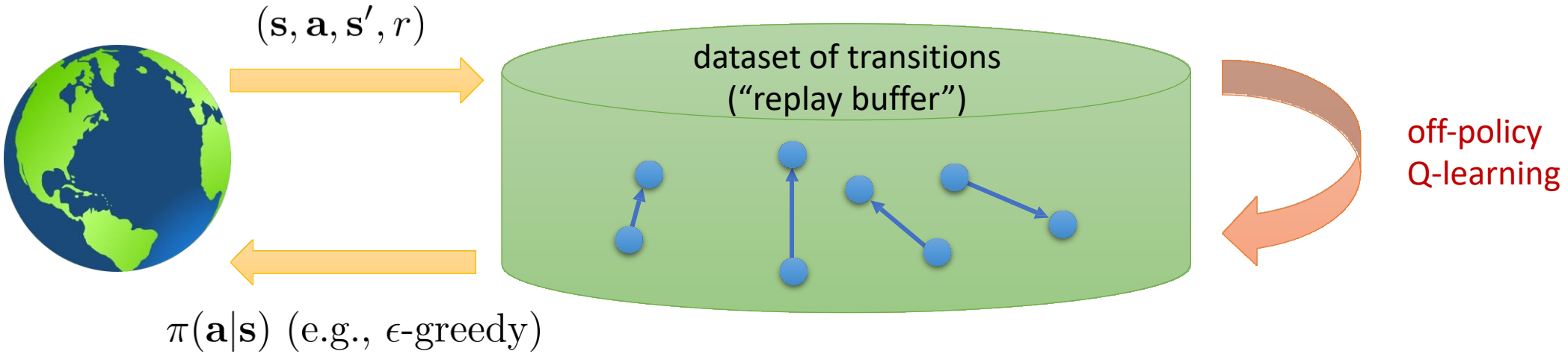
2. set $\mathbf{y}_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'_i} Q_\phi(\mathbf{s}'_i, \mathbf{a}'_i)$
3. set $\phi \leftarrow \arg\min_\phi \frac{1}{2} \sum_i \|Q_\phi(\mathbf{s}_i, \mathbf{a}_i) - \mathbf{y}_i\|^2$



$Q_\phi(\mathbf{s}, \mathbf{a})$

Problem 1): Correlated samples;

Solution: Replay Buffer;



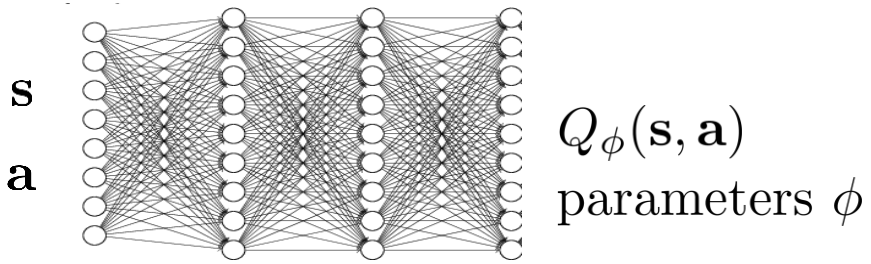
“Max” Trick to Remove Policy Gradient

Problem 2): Q-learning is *not* gradient descent!

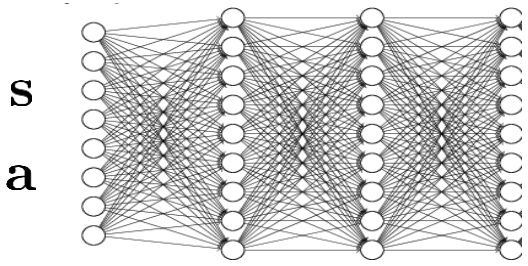
$$\phi \leftarrow \phi - \alpha \frac{dQ_\phi}{d\phi}(\mathbf{s}_i, \mathbf{a}_i)(Q_\phi(\mathbf{s}_i, \mathbf{a}_i) - [r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_\phi(\mathbf{s}'_i, \mathbf{a}'_i)])$$

no gradient through target value

Idea?



$Q_\phi(\mathbf{s}, \mathbf{a})$
parameters ϕ



$Q_{\phi'}$

Target network

Targets don’t change in inner loop! But regression is more stable.

Q-Learning with Replay Buffer and Target Network

1. save target network parameters: $\phi' \leftarrow \phi$

$N\times$ / $K\times$

2. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)\}$ using some policy, add it to $\mathcal{B}$
3. sample a batch $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)$ from $\mathcal{B}$
4. $\phi \leftarrow \phi - \alpha \sum_i \frac{dQ_\phi}{d\phi}(\mathbf{s}_i, \mathbf{a}_i)(Q_\phi(\mathbf{s}_i, \mathbf{a}_i) - [r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}'_i, \mathbf{a}'_i)])$

Supervised Regression

Targets don't change in inner loop!

1. take some action $\mathbf{a}_i$ and observe $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)$, add it to $\mathcal{B}$
2. sample mini-batch $\{\mathbf{s}_j, \mathbf{a}_j, \mathbf{s}'_j, r_j\}$ from $\mathcal{B}$ uniformly
3. compute $y_j = r_j + \gamma \max_{\mathbf{a}'_j} Q_{\phi'}(\mathbf{s}'_j, \mathbf{a}'_j)$ using *target* network $Q_{\phi'}$
4. $\phi \leftarrow \phi - \alpha \sum_j \frac{dQ_\phi}{d\phi}(\mathbf{s}_j, \mathbf{a}_j)(Q_\phi(\mathbf{s}_j, \mathbf{a}_j) - y_j)$
5. update ϕ': copy ϕ every N steps

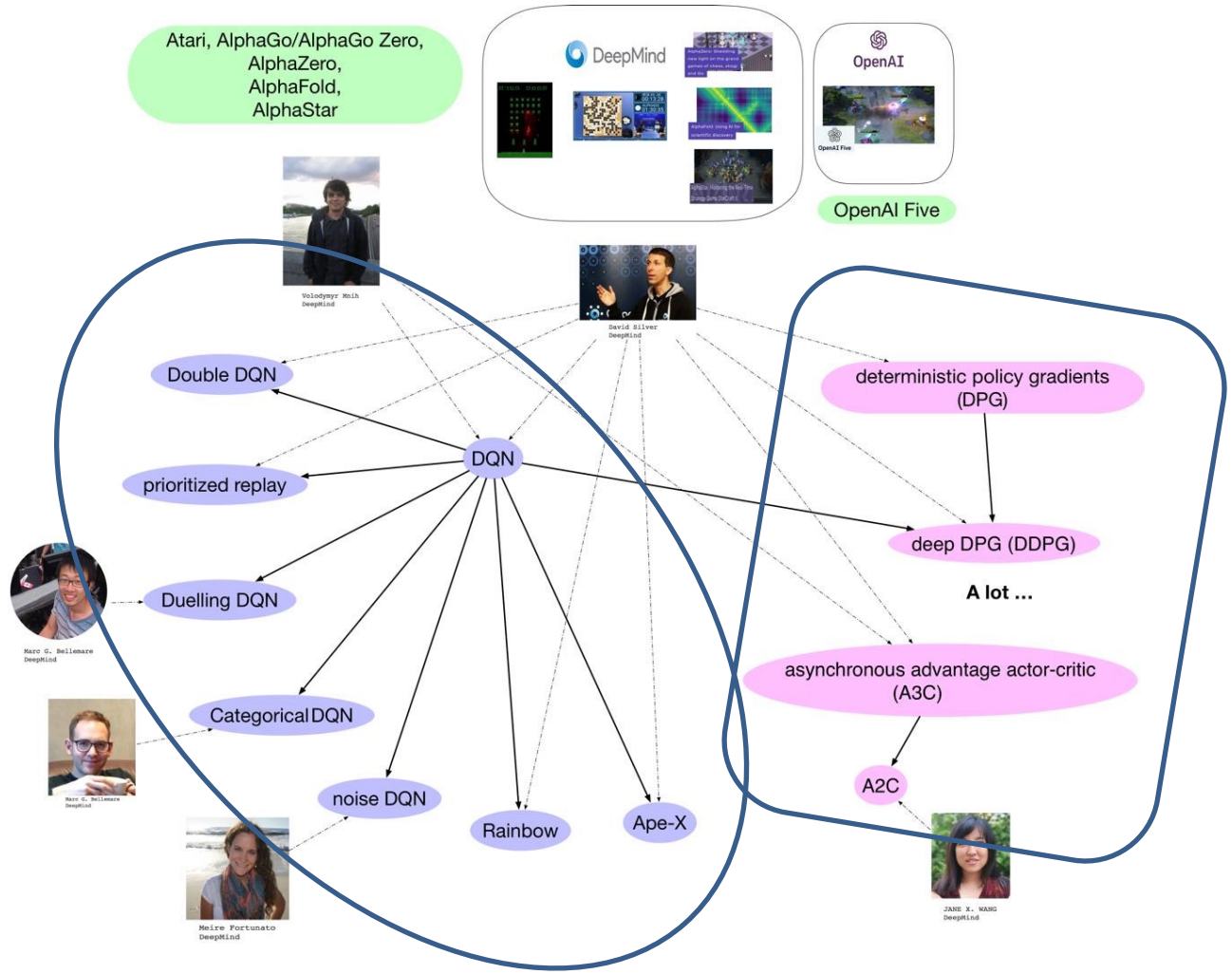
$K = 1$

Target: copy value network!

5. update ϕ': $\phi' \leftarrow \tau\phi' + (1-\tau)\phi$ $\quad\quad$ $\tau = 0.999$ works well

DRL Algorithms

- DQN (Deep Q Network):
  - Use NN to estimate Q;
  - Experience replay and fixed target network for convergence;
  - Variants: Double DQN, Prioritized replay, Dueling DQN, Categorical DQN, Noise DQN, Rainbow
- PG and Actor-Critic:
  - Variants: AC, A2C, A3C and SAC
- DPG and DDPG:
  - Deterministic policy gradients
- TRPO and PPO

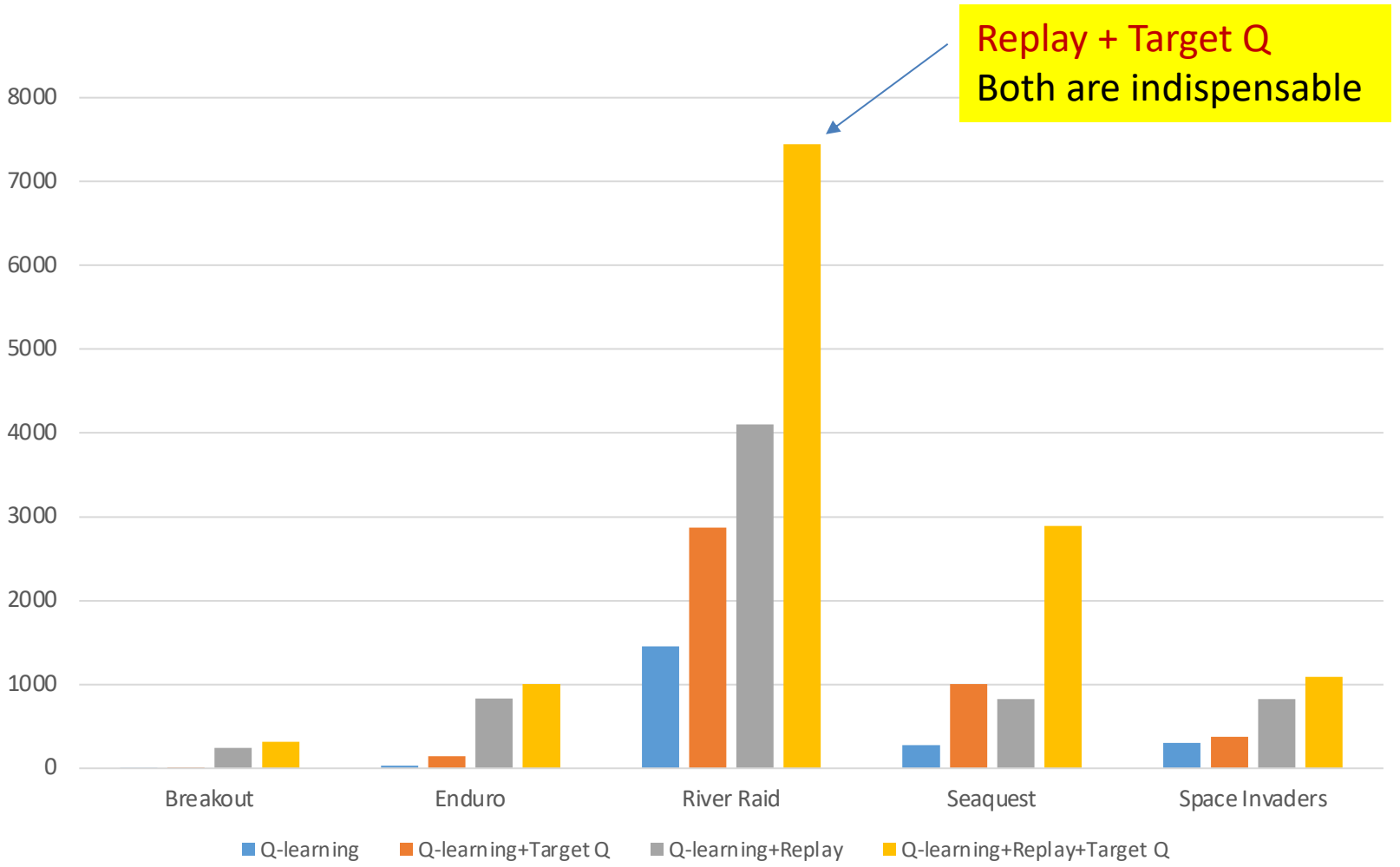


Deep Q Network

- DQN: Use deep NN to compute Q, which is a value-based method
- Before DQN, all attempts failed due to the instability:
  - Unknown reward scale of Q-value, leading to the failure of gradient BP;
  - Strong correlation between continuous-input data or image, leading to easy convergence;
  - Even a fine variation on Q-value causes a huge change on policy (From one end to another).
- Solution from DeepMind
  - Clip rewards or normalize network adaptively to sensible range;
  - Experience Reply (NIPS): store experience (s,a,r,s'); randomly drawn;
  - Freeze target Q-network:

$$loss = \left(r + \gamma \max_{a'} Q(s', a', \mathbf{w}^-) - Q(s, a, \mathbf{w}) \right)^2$$

Deep Q Network



Overestimation of Q-learning in DQN

Overestimation: target value $y_j = r_j + \gamma \max_{\mathbf{a}'_j} Q_{\phi'}(\mathbf{s}'_j, \mathbf{a}'_j)$

this last term is the problem

$Q_{\phi'}(\mathbf{s}', \mathbf{a}')$ is not perfect – it looks "noisy"

hence $\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}')$ *overestimates* the next value!

note that $\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}') = Q_{\phi'}(\mathbf{s}', \arg\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}'))$

value *also* comes from $Q_{\phi'}$ action selected according to $Q_{\phi'}$

Double DQN

note that $\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}') = Q_{\phi'}(\mathbf{s}', \arg\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}'))$

value *also* comes from $Q_{\phi'}$ action selected according to $Q_{\phi'}$

if the noise in these is decorrelated, the problem goes away!

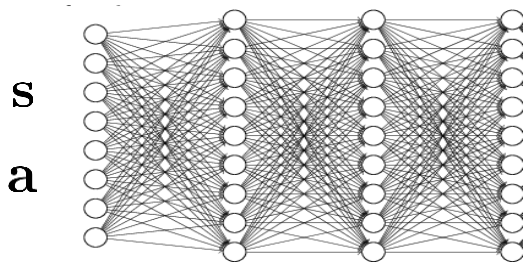
idea: don't use the same network to choose the action and evaluate value!

standard Q-learning: $y = r + \gamma Q_{\phi'}(\mathbf{s}', \arg\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}', \mathbf{a}'))$

double Q-learning: $y = r + \gamma Q_{\phi'}(\mathbf{s}', \arg\max_{\mathbf{a}'} Q_{\phi}(\mathbf{s}', \mathbf{a}'))$

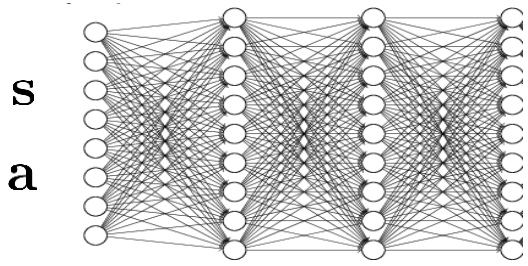
just use current network (not target network) to evaluate action

still use target network to evaluate value!



$Q_\phi(\mathbf{s}, \mathbf{a})$

Current network: action



$Q_{\phi'}$

Target network: value

Multi-Step Returns

Q-learning target: $y_{j,t} = r_{j,t} + \gamma \max_{\mathbf{a}_{j,t+1}} Q_{\phi'}(\mathbf{s}_{j,t+1}, \mathbf{a}_{j,t+1})$

these are the only values that matter if $Q_{\phi'}$ is bad!

these values are important if $Q_{\phi'}$ is good

can we construct multi-step targets, like in actor-critic?

$$y_{j,t} = \sum_{t'=t}^{t+N-1} \gamma^{t-t'} r_{j,t'} + \gamma^N \max_{\mathbf{a}_{j,t+N}} Q_{\phi'}(\mathbf{s}_{j,t+N}, \mathbf{a}_{j,t+N})$$

N-step return estimator

+ less biased target values when Q-values are inaccurate

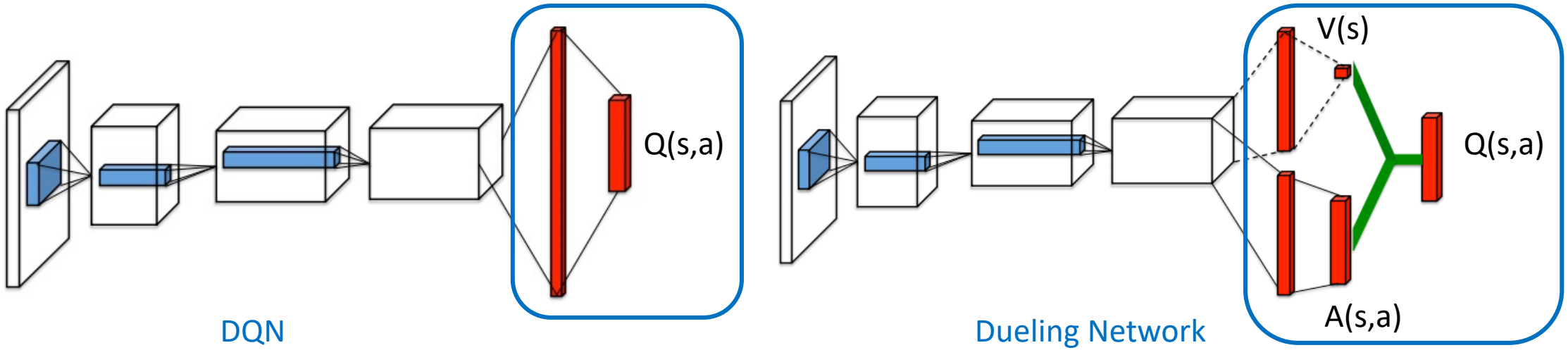
this is supposed to estimate $Q^\pi(\mathbf{s}_{j,t}, \mathbf{a}_{j,t})$ for π

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q_\phi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

Dueling DQN

- Dueling Network: Q=V+A separate state value V and advantage A
  - Value function V measures the value how good it is to be in a particular state s.
  - However, Q function measures the value of choosing a particular action when in this state.
  - Advantage function A obtains a relative measure of the importance of each action.

$$Q(s,a;\theta,\alpha,\beta) = V(s;\theta,\beta) + A(s,a;\theta,\alpha)$$

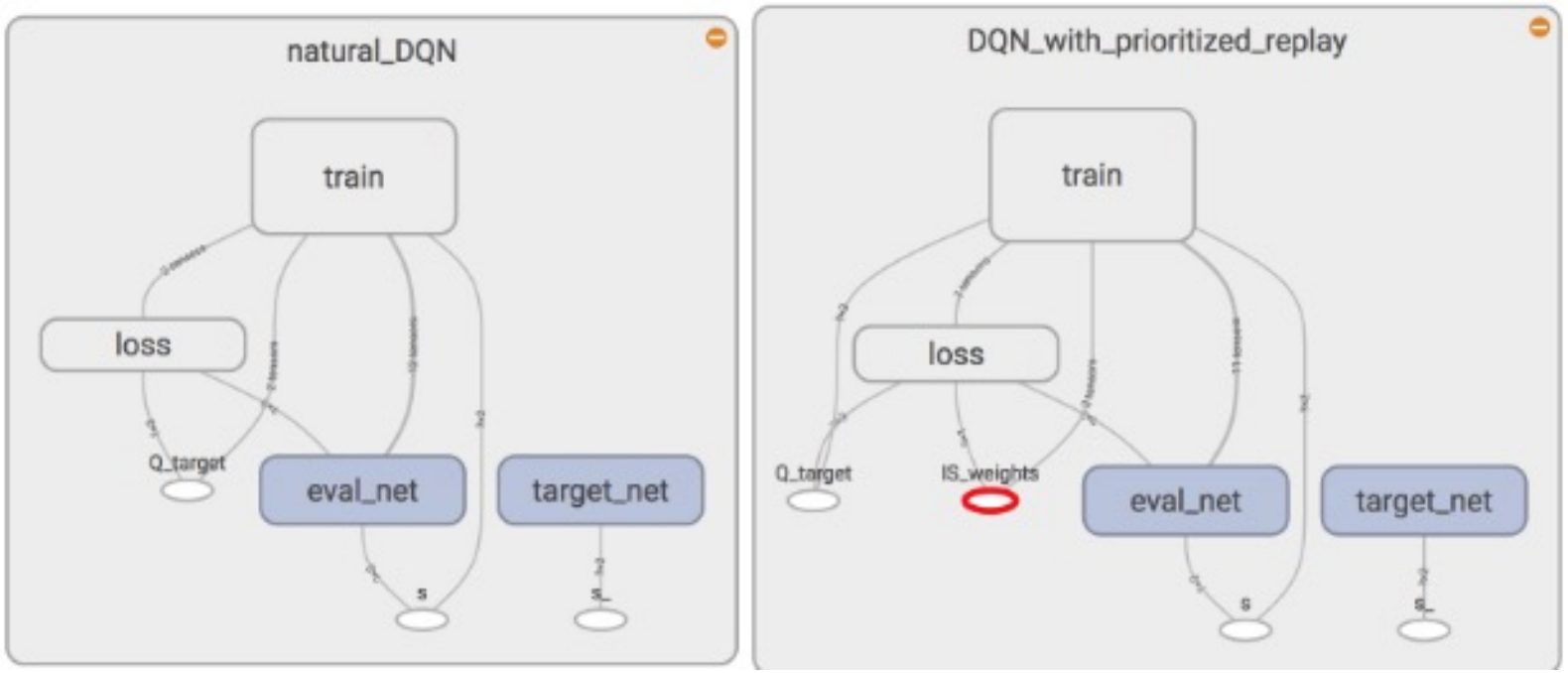


DQN

Dueling Network

Prioritized Replay

- Use priority queue to weight experiences in experience Memory based on their error (surprise) in DQN
- The bigger the TD error, the higher the priority.

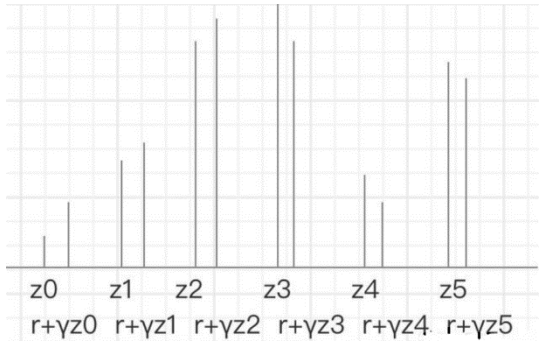


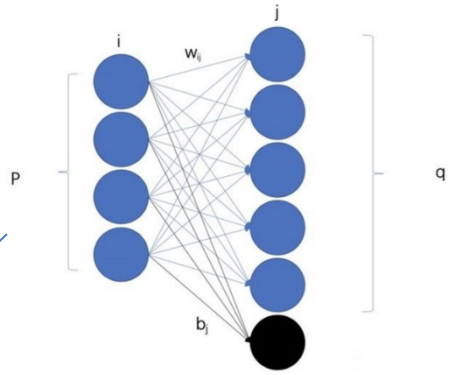
Rainbow

- Rainbow is a model-free, off-policy, value-based and discrete DRL method.
- Rainbow combines all 6 improvements in DQN, including
  - Double Q-learning
  - Multi-step learning
  - Dueling networks
  - Prioritized replay
  - Distributional RL: Q value becomes Q distribution (more stable)

$$r + \gamma \max_a Q(s_{t+1}, a)$$

$$Q(s_{t+1}, a) = \sum_i z_i p_i(s_{t+1}, a)$$





- Noisy Nets

  1) independent Gaussian Noise: add noise on weights and No. is p*(q+1).

  2) Factorized Gaussian noise: add noise on neurons and No. is p+q

Soft Actor-Critic

- Standard RL maximizes the expected sum of rewards:

$$\sum_t \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} \left[r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

- SAC favors stochastic policies by augmenting the objective with the expected entropy of the policy:

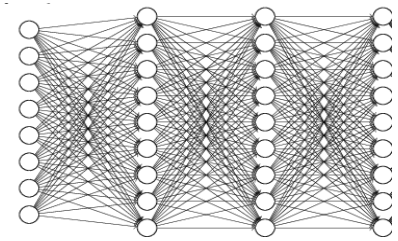
$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} \left[r(\mathbf{s}_t, \mathbf{a}_t) + \alpha \mathcal{H}(\pi(\cdot | \mathbf{s}_t)) \right]$$

- Soft state value function:

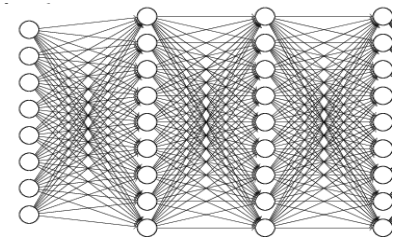
$$V(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t \sim \pi} \left[Q(\mathbf{s}_t, \mathbf{a}_t) - \log \pi(\mathbf{a}_t | \mathbf{s}_t) \right]$$

Soft Actor-Critic

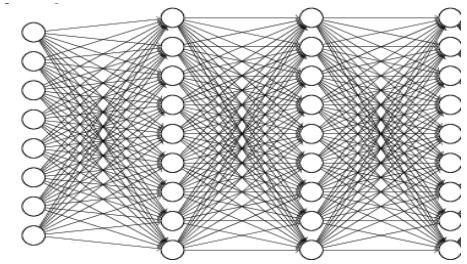
Value Network



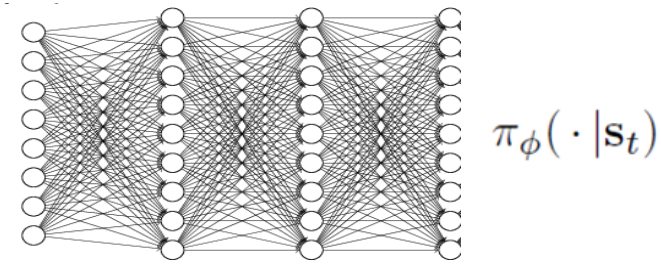
Value Target



Q Network



Policy Network



- Soft value function V (MSE):

$$J_V(\psi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\frac{1}{2} \left(V_\psi(\mathbf{s}_t) - \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} \left[Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \log \pi_\phi(\mathbf{a}_t|\mathbf{s}_t) \right] \right)^2 \right]$$

$$\hat{\nabla}_\psi J_V(\psi) = \nabla_\psi V_\psi(\mathbf{s}_t) \left(V_\psi(\mathbf{s}_t) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t) + \log \pi_\phi(\mathbf{a}_t|\mathbf{s}_t) \right)$$

Soft Actor-Critic

- Soft Q-function parameters Q (MSE):

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}} \left[\frac{1}{2} \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \hat{Q}(\mathbf{s}_t, \mathbf{a}_t) \right)^2 \right]$$

$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p} \left[V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right]$$

$$\hat{\nabla}_\theta J_Q(\theta) = \nabla_\theta Q_\theta(\mathbf{a}_t, \mathbf{s}_t) \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - r(\mathbf{s}_t, \mathbf{a}_t) - \gamma V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right)$$

- Policy parameters learned by minimizing expected KL-divergence:

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\mathrm{D}_{\mathrm{KL}} \left(\pi_\phi(\cdot | \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(Q_\theta(\mathbf{s}_t, \cdot)\right)}{Z_\theta(\mathbf{s}_t)} \right) \right]$$

- Target value network (for overestimate): moving average of value network weight

$$\bar{\psi} \leftarrow \tau \psi + (1 - \tau) \bar{\psi}$$

Soft Actor-Critic

Algorithm 1 Soft Actor-Critic

Initialize parameter vectors ψ, $\bar{\psi}$, θ, ϕ.
for each iteration **do**
  for each environment step **do**
    $\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$
    $\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_V \hat{\nabla}_\psi J_V(\psi)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$
    $\phi \leftarrow \phi - \lambda_\pi \nabla_\phi J_\pi(\phi)$
    $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
  end for
end for

Update value V

Update Q

Update Policy

Update target value network

use the minimum of Q-functions for the value gradient

Q-learning with continuous actions

What's the problem with continuous actions?

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q_\phi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

this max

$$\text{target value } y_j = r_j + \gamma \max_{\mathbf{a}'_j} Q_{\phi'}(\mathbf{s}'_j, \mathbf{a}'_j)$$

this max particularly problematic

How do we perform the max?

DDPG-Learn an Approximate Maximizer

$\max_{\mathbf{a}} Q_\phi(\mathbf{s}, \mathbf{a}) = Q_\phi(\mathbf{s}, \arg\max_{\mathbf{a}} Q_\phi(\mathbf{s}, \mathbf{a}))$

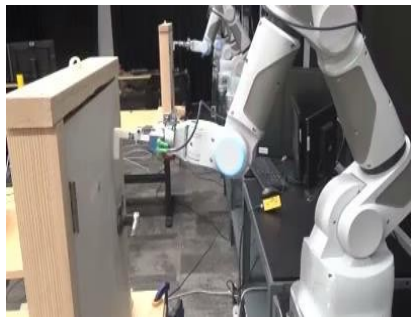
idea: train another network $\mu_\theta(\mathbf{s})$ such that $\mu_\theta(\mathbf{s}) \approx \arg\max_{\mathbf{a}} Q_\phi(\mathbf{s}, \mathbf{a})$

how? just solve $\theta \leftarrow \arg\max_\theta Q_\phi(\mathbf{s}, \mu_\theta(\mathbf{s}))$ $\quad \frac{dQ_\phi}{d\theta} = \frac{d\mathbf{a}}{d\theta}\frac{dQ_\phi}{d\mathbf{a}}$

new target $y_j = r_j + \gamma Q_{\phi'}(\mathbf{s}'_j, \mu_\theta(\mathbf{s}'_j)) \approx r_j + \gamma Q_{\phi'}(\mathbf{s}'_j, \arg\max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}'_j, \mathbf{a}'_j))$

DDPG:

1. take some action $\mathbf{a}_i$ and observe $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)$, add it to $\mathcal{B}$
2. sample mini-batch $\{\mathbf{s}_j, \mathbf{a}_j, \mathbf{s}'_j, r_j\}$ from $\mathcal{B}$ uniformly
3. compute $y_j = r_j + \gamma Q_{\phi'}(\mathbf{s}'_j, \mu_{\theta'}(\mathbf{s}'_j))$ using *target* nets $Q_{\phi'}$ and $\mu_{\theta'}$
4. $\phi \leftarrow \phi - \alpha \sum_j \frac{dQ_\phi}{d\phi}(\mathbf{s}_j, \mathbf{a}_j)(Q_\phi(\mathbf{s}_j, \mathbf{a}_j) - y_j)$ Value Network
5. $\theta \leftarrow \theta + \beta \sum_j \frac{d\mu}{d\theta}(\mathbf{s}_j)\frac{dQ_\phi}{d\mathbf{a}}(\mathbf{s}_j, \mu(\mathbf{s}_j))$ Policy Network; deterministic
6. update ϕ' and θ' (e.g., Polyak averaging)



Trust Region Policy Optimization (TRPO)

Recall:

REINFORCE algorithm:

1. sample $\{\tau^i\}$ from $\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ (run the policy)
2. $\nabla_\theta J(\theta) \approx \sum_i \left(\sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^i|\mathbf{s}_t^i)\right)\left(\sum_t r(\mathbf{s}_t^i, \mathbf{a}_t^i)\right)$
3. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

Problem: *unstable!*

Bad α may cause terrible policy π_θ!

Question: *How to make policy monotonic improved? (always cause better policy?)*

$$\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})\left(\hat{Q}_{i,t} - \mathbb{E}_a \hat{Q}_{i,t}\right)$$

$\hat{A}_{i,t}$

$$\text{s.t.} \quad \underbrace{D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \leq \delta.}_{\text{“Trust Region”}}$$

$$\hat{\mathbb{E}}_t\left[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\hat{A}_t\right]$$

$$\hat{\mathbb{E}}_t\left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\hat{A}_t\right]$$

Proximal Policy Optimization (PPO)

Off-policy Policy Gradient

$$\nabla_{\theta'} J(\theta') = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \frac{\pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})}{\pi_{\theta}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}) \hat{Q}_{i,t}$$

Variance Reducing 2

$$\nabla_{\theta'} J(\theta') = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \frac{\pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})}{\pi_{\theta}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}) \underbrace{\left(\hat{Q}_{i,t} - \mathbb{E}\hat{Q}_{i,t}\right)}$$

"advantage" $\hat{A}_{i,t}$

How to introduce trust region efficiently? ***CLIP:*** $\text{clip}(x, l, u) = \min(\max(x, l), u)$

$$\nabla_{\theta'} J(\theta') = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \underbrace{\text{clip}\left(\frac{\pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})}{\pi_{\theta}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})}, 1-\epsilon, 1+\epsilon\right)} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}) \left(\hat{Q}_{i,t} - \mathbb{E}\hat{Q}_{i,t}\right)$$

"Trust Region by Clip"

TRPO and PPO

Policy Gradient Methods

$$L^{PG}(\theta) = \hat{\mathbb{E}}_t\left[\log \pi_\theta(a_t \mid s_t)\hat{A}_t\right]$$

$$\hat{g} = \hat{\mathbb{E}}_t\left[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\hat{A}_t\right]$$

Trust Region Methods (TRPO)

$$\begin{aligned} \underset{\theta}{\text{maximize}} \quad & \hat{\mathbb{E}}_t\left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\hat{A}_t\right] \\ \text{subject to} \quad & \hat{\mathbb{E}}_t[\mathrm{KL}[\pi_{\theta_{\text{old}}}(\cdot \mid s_t), \pi_\theta(\cdot \mid s_t)]] \leq \delta. \end{aligned}$$

Proximal Policy Optimization (PPO)

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right]$$

Model-based DRL

model-based reinforcement learning version 1.0:

1. run base policy $\pi_0(\mathbf{a}_t|\mathbf{s}_t)$ (e.g., random policy) to collect $\mathcal{D} = \{(\mathbf{s}, \mathbf{a}, \mathbf{s}')_i\}$
2. learn dynamics model $f(\mathbf{s}, \mathbf{a})$ to minimize $\sum_i \|f(\mathbf{s}_i, \mathbf{a}_i) - \mathbf{s}'_i\|^2$ → Model
3. plan through $f(\mathbf{s}, \mathbf{a})$ to choose actions → Planning
4. execute those actions and add the resulting data $\{(\mathbf{s}, \mathbf{a}, \mathbf{s}')_j\}$ to $\mathcal{D}$

model-based reinforcement learning version 1.5:

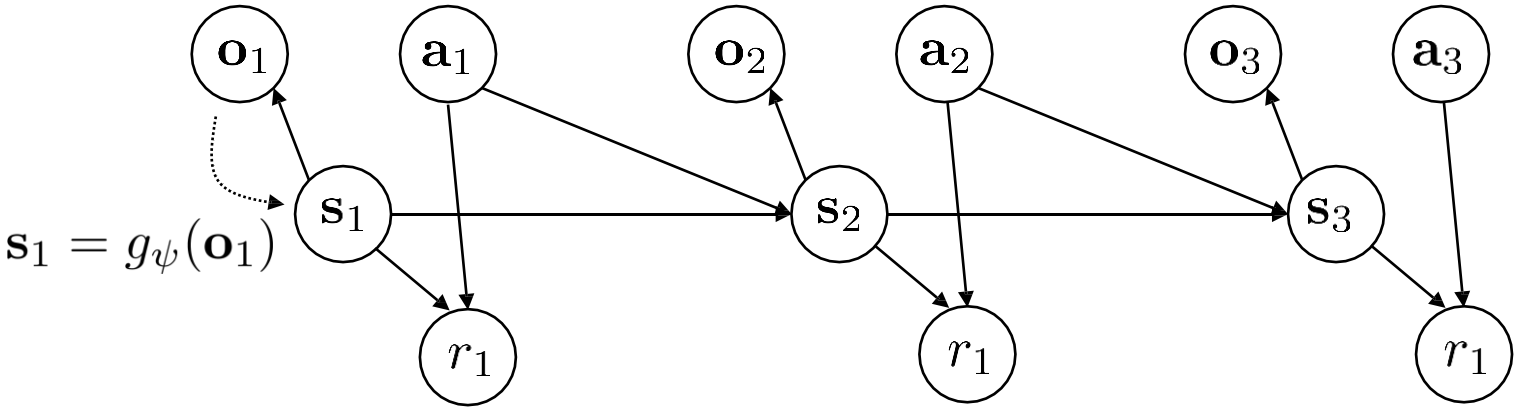
1. run base policy $\pi_0(\mathbf{a}_t|\mathbf{s}_t)$ (e.g., random policy) to collect $\mathcal{D} = \{(\mathbf{s}, \mathbf{a}, \mathbf{s}')_i\}$
2. learn dynamics model $f(\mathbf{s}, \mathbf{a})$ to minimize $\sum_i \|f(\mathbf{s}_i, \mathbf{a}_i) - \mathbf{s}'_i\|^2$ → Model
3. plan through $f(\mathbf{s}, \mathbf{a})$ to choose actions → Planning

Replanning

4. execute the first planned action, observe resulting state $\mathbf{s}'$ (MPC)
5. append $(\mathbf{s}, \mathbf{a}, \mathbf{s}')$ to dataset $\mathcal{D}$

every N steps

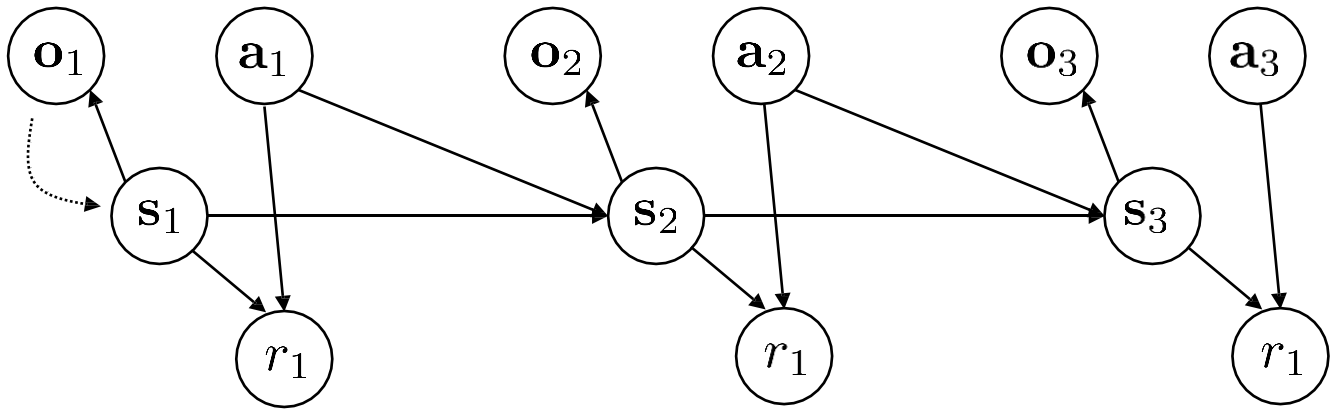
Model-based DRL with Latent Space Models



$$\max_{\phi,\psi} \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \underbrace{\log p_\phi(g_\psi(\mathbf{o}_{t+1,i}) | g_\psi(\mathbf{o}_{t,i}), \mathbf{a}_{t,i})}_{\text{Latent Space Dynamics}} + \underbrace{\log p_\phi(\mathbf{o}_{t,i} | g_\psi(\mathbf{o}_{t,i}))}_{\text{Image Reconstruction}} + \underbrace{\log p_\phi(r_{t,i} | g_\psi(\mathbf{o}_{t,i}))}_{\text{Reward Model}}$$

Many practical methods consider a Stochastic Encoder for Model Uncertainty.

Model-based DRL with Latent Space Models



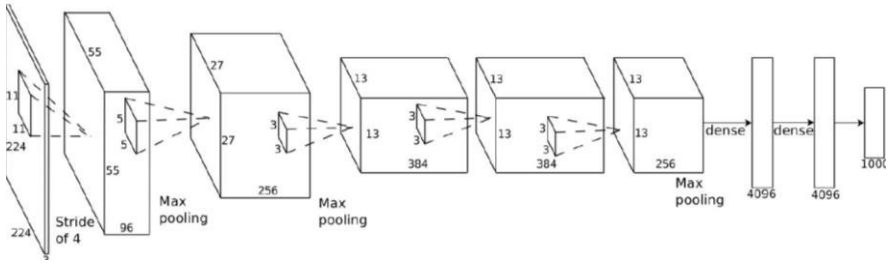
model-based reinforcement learning with latent state:

every N steps

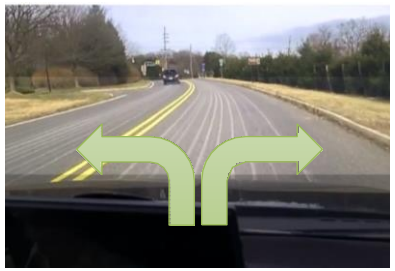
1. run base policy $\pi_0(\mathbf{a}_t|\mathbf{o}_t)$ (e.g., random policy) to collect $\mathcal{D} = \{(\mathbf{o}, \mathbf{a}, \mathbf{o}')_i\}$
2. learn $p_\phi(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$, $p_\phi(r_t|\mathbf{s}_t)$, $p(\mathbf{o}_t|\mathbf{s}_t)$, $g_\psi(\mathbf{o}_t)$
3. plan through the model to choose actions
4. execute the first planned action, observe resulting $\mathbf{o}'$ (MPC)
5. append $(\mathbf{o}, \mathbf{a}, \mathbf{o}')$ to dataset $\mathcal{D}$

Imitation Learning

$\mathbf{o}_t$



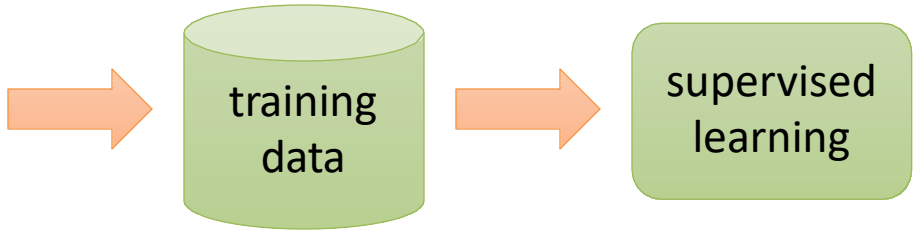
$\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$



$\mathbf{a}_t$

$\mathbf{o}_t$

$\mathbf{a}_t$

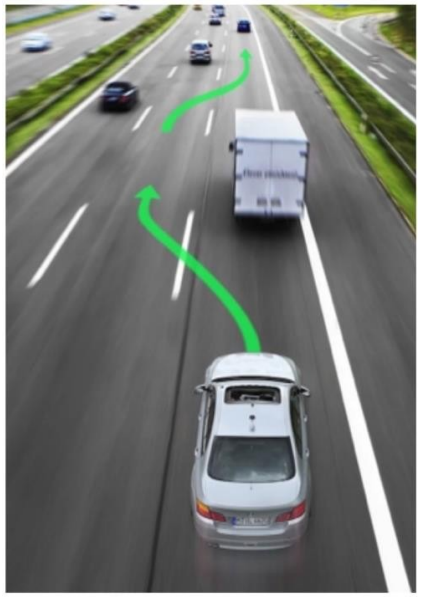


$\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$

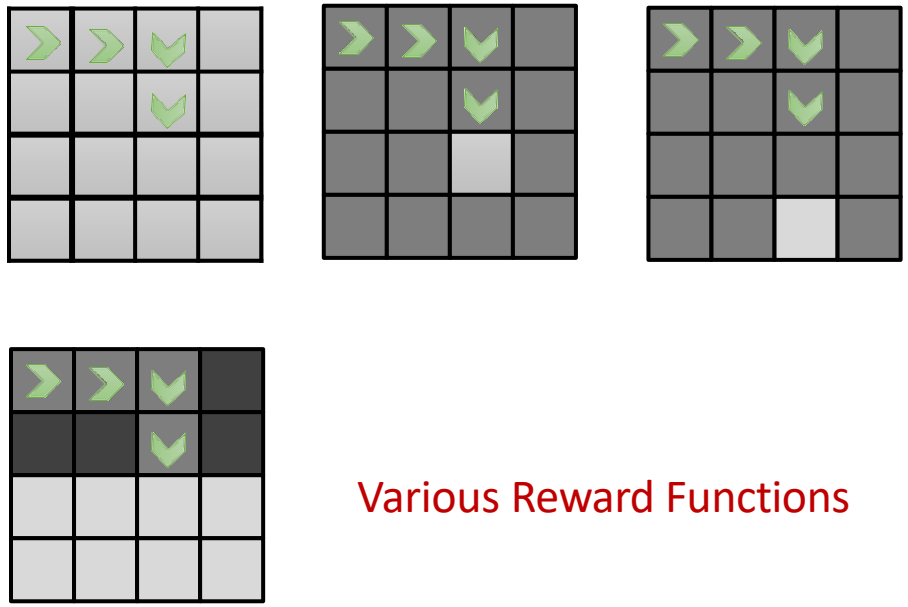
Behavioral Cloning

Inverse Reinforcement Learning (IRL)

Infer reward functions from demonstrations



$r(\mathbf{s}, \mathbf{a})$



Various Reward Functions

- Underspecified problem
- Many reward functions can explain the same behavior

Inverse Reinforcement Learning (IRL)

"forward" reinforcement learning

given:

states $\mathbf{s} \in \mathcal{S}$, actions $\mathbf{a} \in \mathcal{A}$

(sometimes) transitions $p(\mathbf{s}'|\mathbf{s}, \mathbf{a})$

reward function $r(\mathbf{s}, \mathbf{a})$

learn $\pi^\star(\mathbf{a}|\mathbf{s})$

Inverse reinforcement learning

given:

states $\mathbf{s} \in \mathcal{S}$, actions $\mathbf{a} \in \mathcal{A}$

(sometimes) transitions $p(\mathbf{s}'|\mathbf{s}, \mathbf{a})$

samples $\{\tau_i\}$ sampled from $\pi^\star(\tau)$

learn $r_\psi(\mathbf{s}, \mathbf{a})$

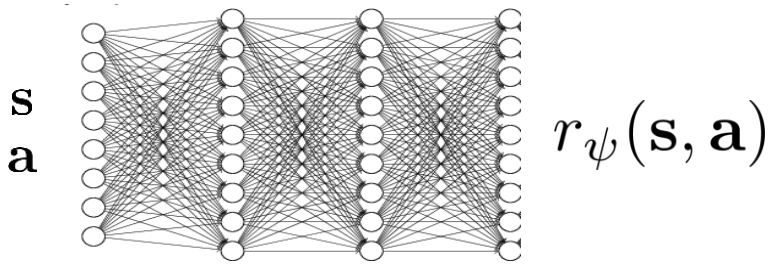
Reward Parameters

...and then use it to learn $\pi^\star(\mathbf{a}|\mathbf{s})$

linear reward function:

$r_\psi(\mathbf{s}, \mathbf{a}) = \sum_i \psi_i f_i(\mathbf{s}, \mathbf{a}) = \psi^T \mathbf{f}(\mathbf{s}, \mathbf{a})$

neural net reward function:



$r_\psi(\mathbf{s}, \mathbf{a})$

Learn Optimality Variable

given:

samples $\{\tau_i\}$ sampled from $\pi^\star(\tau)$

$$p(\tau|\mathcal{O}_{1:T}, \psi) \propto \cancel{p(\tau)} \exp\left(\sum_t r_\psi(\mathbf{s}_t, \mathbf{a}_t)\right)$$

can ignore (independent of ψ)

$$Z = \int p(\tau) \exp(r_\psi(\tau)) d\tau$$

maximum likelihood learning:

$$\max_\psi \frac{1}{N} \sum_{i=1}^{N} \log p(\tau_i|\mathcal{O}_{1:T}, \psi) = \max_\psi \frac{1}{N} \sum_{i=1}^{N} r_\psi(\tau_i) - \log Z$$

partition function

$$\nabla_\psi \mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \nabla_\psi r_\psi(\tau_i) - \underbrace{\frac{1}{Z} \int p(\tau) \exp(r_\psi(\tau)) \nabla_\psi r_\psi(\tau) d\tau}_{p(\tau|\mathcal{O}_{1:T}, \psi)}$$

$$\nabla_\psi \mathcal{L} = E_{\tau \sim \pi^\star(\tau)}[\nabla_\psi r_\psi(\tau_i)] - E_{\tau \sim p(\tau|\mathcal{O}_{1:T}, \psi)}[\nabla_\psi r_\psi(\tau)]$$

estimate with expert samples

soft optimal policy under current reward

Estimation of Expectation

$$\nabla_\psi \mathcal{L} = E_{\tau\sim\pi^\star(\tau)}[\nabla_\psi r_\psi(\tau_i)] - \underbrace{E_{\tau\sim p(\tau|\mathcal{O}_{1:T},\psi)}[\nabla_\psi r_\psi(\tau)]}$$

$$E_{\tau\sim p(\tau|\mathcal{O}_{1:T},\psi)}\left[\nabla_\psi \sum_{t=1}^{T} r_\psi(\mathbf{s}_t,\mathbf{a}_t)\right]$$

$$= \sum_{t=1}^{T} \underbrace{E_{(\mathbf{s}_t,\mathbf{a}_t)\sim p(\mathbf{s}_t,\mathbf{a}_t|\mathcal{O}_{1:T},\psi)}}\,[\nabla_\psi r_\psi(\mathbf{s}_t,\mathbf{a}_t)]$$

$$p(\mathbf{a}_t|\mathbf{s}_t,\mathcal{O}_{1:T},\psi)p(\mathbf{s}_t|\mathcal{O}_{1:T},\psi)$$

backward message

$\beta_t(\mathbf{s}_t,\mathbf{a}_t) = p(\mathcal{O}_{t:T}|\mathbf{s}_t,\mathbf{a}_t)$

$$= \frac{\beta(\mathbf{s}_t,\mathbf{a}_t)}{\beta(\mathbf{s}_t)}$$

$$\propto \alpha(\mathbf{s}_t)\beta(\mathbf{s}_t)$$

forward message $\alpha_t(\mathbf{s}_t) = p(\mathbf{s}_t|\mathcal{O}_{1:t-1})$

$$p(\mathbf{a}_t|\mathbf{s}_t,\mathcal{O}_{1:T},\psi)p(\mathbf{s}_t|\mathcal{O}_{1:T},\psi) \propto \beta(\mathbf{s}_t,\mathbf{a}_t)\alpha(\mathbf{s}_t)$$

IRL Algorithm: MaxEnt

let $\mu_t(\mathbf{s}_t, \mathbf{a}_t) \propto \beta(\mathbf{s}_t, \mathbf{a}_t)\alpha(\mathbf{s}_t)$

$$\nabla_\psi \mathcal{L} = E_{\tau \sim \pi^\star(\tau)}[\nabla_\psi r_\psi(\tau_i)] - E_{\tau \sim p(\tau|\mathcal{O}_{1:T},\psi)}[\nabla_\psi r_\psi(\tau)]$$

$$\sum_{t=1}^{T} \int \int \mu_t(\mathbf{s}_t, \mathbf{a}_t) \nabla_\psi r_\psi(\mathbf{s}_t, \mathbf{a}_t) d\mathbf{s}_t d\mathbf{a}_t = \sum_{t=1}^{T} \vec{\mu}_t^T \nabla_\psi \vec{r}_\psi$$

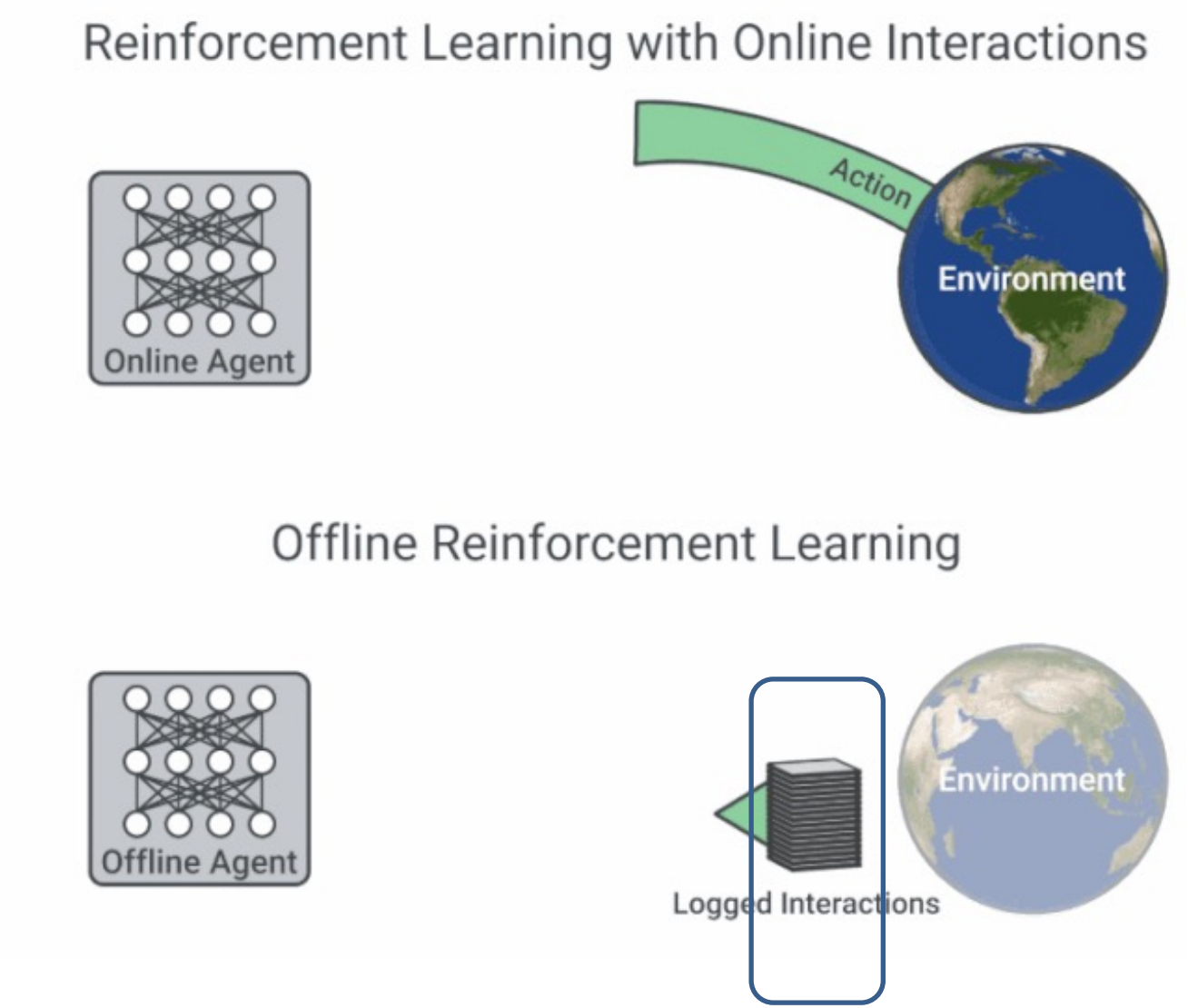
state-action visitation probability for each $(\mathbf{s}_t, \mathbf{a}_t)$

Visitation Frequency

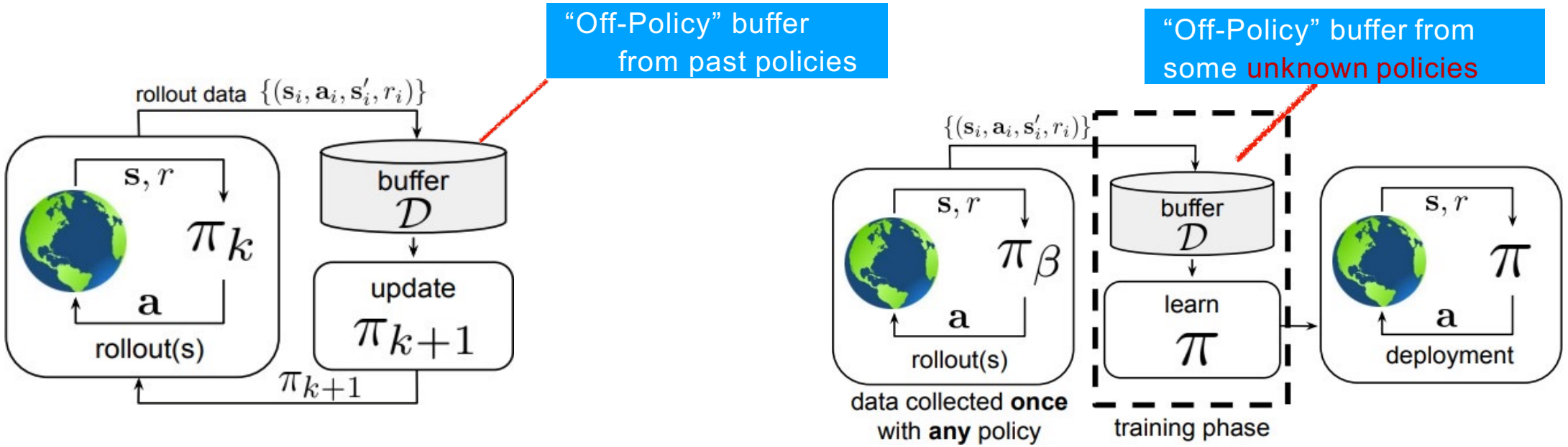
MaxEnt:

1. Given ψ, compute backward message $\beta(\mathbf{s}_t, \mathbf{a}_t)$
2. Given ψ, compute forward message $\alpha(\mathbf{s}_t)$
3. Compute $\mu_t(\mathbf{s}_t, \mathbf{a}_t) \propto \beta(\mathbf{s}_t, \mathbf{a}_t)\alpha(\mathbf{s}_t)$
4. Evaluate $\nabla_\psi \mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} \nabla_\psi r_\psi(\mathbf{s}_{i,t}, \mathbf{a}_{i,t}) - \sum_{t=1}^{T} \int \int \mu_t(\mathbf{s}_t, \mathbf{a}_t) \nabla_\psi r_\psi(\mathbf{s}_t, \mathbf{a}_t) d\mathbf{s}_t d\mathbf{a}_t$
5. $\psi \leftarrow \psi + \eta \nabla_\psi \mathcal{L}$

Offline Reinforcement Learning



Off-policy and Offline DRL



- Off-Policy DRL Algorithms
- Offline DRL Algorithms

Offline Reinforcement Learning

Supervised Learning

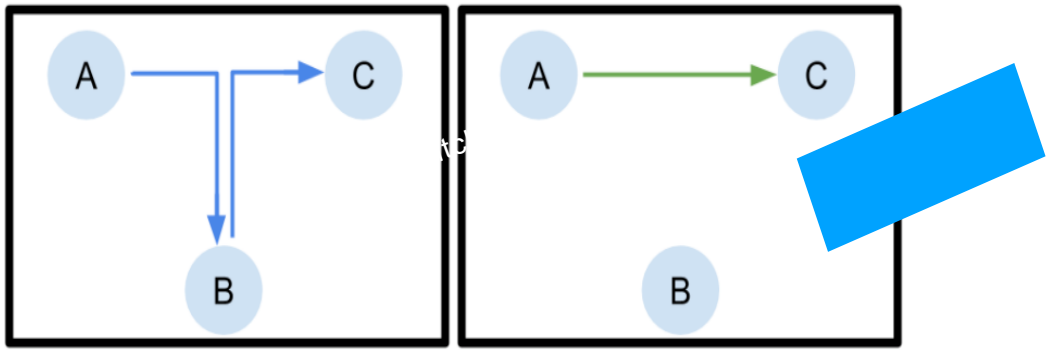
Can do as good as the dataset!

Dog?

Cat?

Offline Reinforcement Learning

Can do **better** than the dataset!



Can show that Q-learning recovers optimal policy from random data.

Conservative Q-Learning (CQL)

- Conservative Q-learning (CQL): aims to address these limitations by learning a conservative Q-function such that the expected value of a policy under this Q-function lower-bounds its true value.

- To prevent overestimation: learn a conservative, lower-bound Q-function by additionally minimizing Q-values alongside Bellman error objective.

Kumar, Zhou, Tucker, Levine. Conservative Q-Learning for Offline RL. NeurIPS 2020.

Conservative Q-Learning (CQL)

- Policy Evaluation:

Minimize big Q-values

$$\hat{Q}^{k+1} \leftarrow \arg\min_{Q} \; \alpha \, \mathbb{E}_{\mathbf{s}\sim\mathcal{D}, \mathbf{a}\sim\mu(\mathbf{a}|\mathbf{s})} [Q(\mathbf{s},\mathbf{a})] + \frac{1}{2} \mathbb{E}_{\mathbf{s},\mathbf{a}\sim\mathcal{D}} \left[\left(Q(\mathbf{s},\mathbf{a}) - \hat{\mathcal{B}}^{\pi} \hat{Q}^{k}(\mathbf{s},\mathbf{a}) \right)^2 \right]$$

- Furthermore, improve the bound by introducing an additional Q-value maximization term.

Minimize big Q-values

Maximize Data Q-values

$$\hat{Q}^{k+1} \leftarrow \arg\min_{Q} \; \alpha \cdot \left(\mathbb{E}_{\mathbf{s}\sim\mathcal{D}} \mathbf{a}\sim\mu(\mathbf{a}|\mathbf{s}) [Q(\mathbf{s},\mathbf{a})] - \mathbb{E}_{\mathbf{s}\sim\mathcal{D}} \mathbf{a}\sim\hat{\pi}_{\beta}(\mathbf{a}|\mathbf{s}) [Q(\mathbf{s},\mathbf{a})] \right)$$
$$+ \frac{1}{2} \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}'\sim\mathcal{D}} \left[\left(Q(\mathbf{s},\mathbf{a}) - \hat{\mathcal{B}}^{\pi} \hat{Q}^{k}(\mathbf{s},\mathbf{a}) \right)^2 \right]$$

Conservative Q-Learning (CQL)

- How should we utilize this for policy optimization?
  - ➢ Alternate between performing full off-policy evaluation for each policy iterate, and one-step of policy improvement.

Minimize big Q-values

Maximize Data Q-values

$$\min_{Q} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{s}\sim\mathcal{D},\mathbf{a}\sim\mu(\mathbf{a}|\mathbf{s})} [Q(\mathbf{s},\mathbf{a})] - \mathbb{E}_{\mathbf{s}\sim\mathcal{D},\mathbf{a}\sim\hat{\pi}_\beta(\mathbf{a}|\mathbf{s})} [Q(\mathbf{s},\mathbf{a})] \right)$$
$$+ \frac{1}{2} \mathbb{E}_{\mathbf{s},\mathbf{a},\mathbf{s}'\sim\mathcal{D}} \left[\left(Q(\mathbf{s},\mathbf{a}) - \hat{\mathcal{B}}^{\pi_k} \hat{Q}^k(\mathbf{s},\mathbf{a}) \right)^2 \right] + \mathcal{R}(\mu) \quad (\mathrm{CQL}(\mathcal{R}))$$

Standard Bellman Error

Regularization

CQL Algorithm:

1. Learn $\hat{Q}^\pi_{\mathrm{CQL}}$ using offline data $\mathcal{D}$.
2. Optimize policy w.r.t. $\hat{Q}^\pi_{\mathrm{CQL}}$: $\pi \leftarrow \arg\max_\pi \mathbb{E}_\pi[\hat{Q}^\pi_{\mathrm{CQL}}]$.

Model-based Offline Policy Optimization (MOPO)

- Standard model-based methods: designed for the online setting, do NOT provide an explicit mechanism to avoid the distributional shift issue.

- MOPO: modify the existing model-based RL by applying them with rewards artificially penalized by the uncertainty of the dynamics.

Model-based Offline Policy Optimization (MOPO)

- MOPO: modify the existing model-based RL by considering such rewards artificially penalized by the uncertainty of the dynamics.

Algorithm 1 Framework for Model-based Offline Policy Optimization (MOPO) with Reward Penalty

Require: Dynamics model $\widehat{T}$ with admissible error estimator $u(s,a)$; constant λ.

1: Define $\tilde{r}(s,a) = r(s,a) - \lambda u(s,a)$. Let $\widetilde{M}$ be the MDP with dynamics $\widehat{T}$ and reward $\tilde{r}$.

2: Run any RL algorithm on $\widetilde{M}$ until convergence to obtain $\hat{\pi} = \text{argmax}_{\pi} \eta_{\widetilde{M}}(\pi)$

- Maximum standard deviation of the learned models in the ensemble:

$$u(s,a) = \max_{i=1}^{N} \|\Sigma_{\phi}^{i}(s,a)\|_{\text{F}}$$

$$\tilde{r}(s,a) = \hat{r}(s,a) - \lambda \max_{i=1,\ldots,N} \|\Sigma_{\phi}^{i}(s,a)\|_{\text{F}}$$

Model-based Offline Policy Optimization (MOPO)

Algorithm 1 Framework for Model-based Offline Policy Optimization (MOPO) with Reward Penalty

Require: Dynamics model $\widehat{T}$ with admissible error estimator $u(s,a)$; constant λ.

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2: Run any RL algorithm on $\widetilde{M}$ until convergence to obtain $\hat{\pi} = \text{argmax}_{\pi} \eta_{\widetilde{M}}(\pi)$

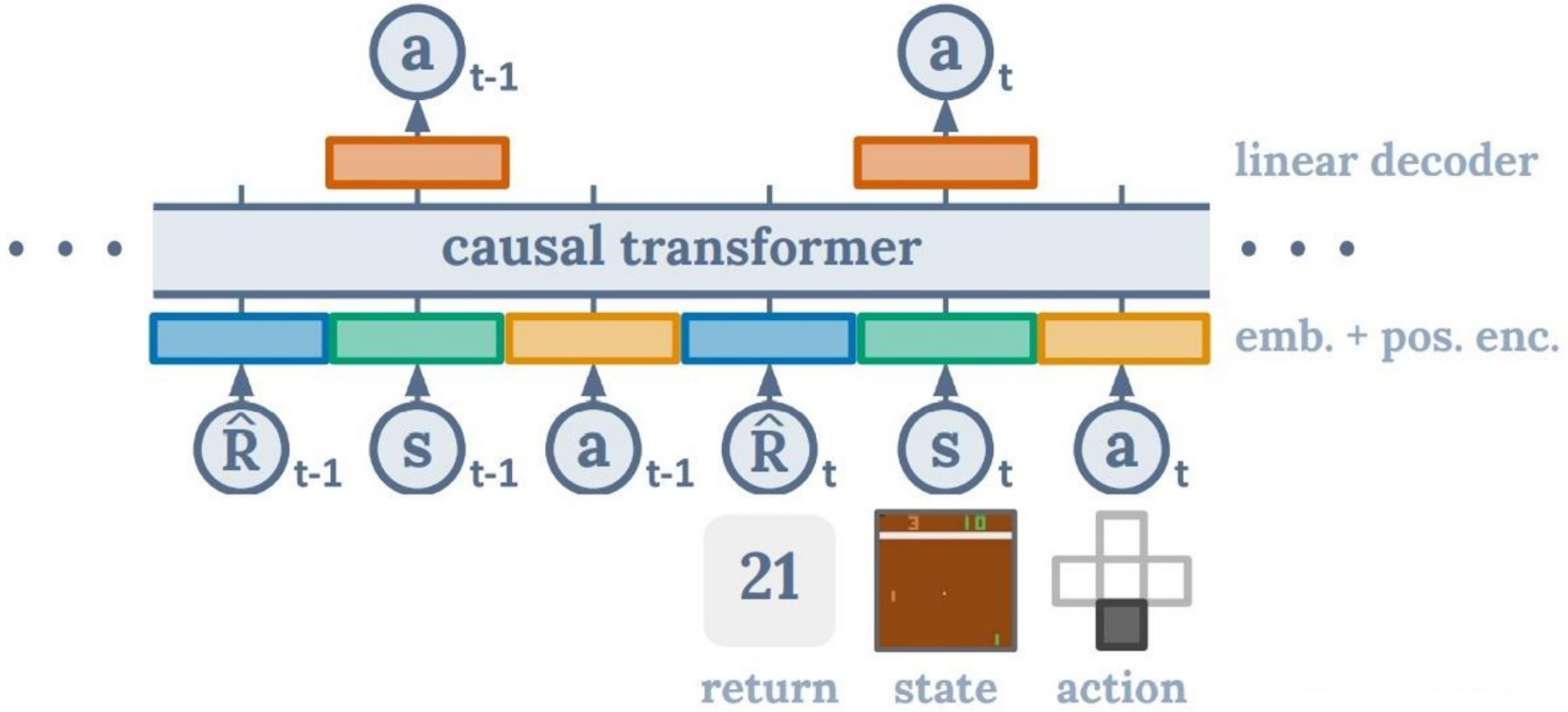
- Model the dynamics using a neural network that outputs a Gaussian distribution over the next state and reward:

$$\widehat{T}_{\theta,\phi}(s_{t+1}, r | s_t, a_t) = \mathcal{N}(\mu_\theta(s_t, a_t), \Sigma_\phi(s_t, a_t))$$

- We learn an ensemble of N dynamics models, with each model trained independently via maximum likelihood.

$$\{\widehat{T}^i_{\theta,\phi} = \mathcal{N}(\mu^i_\theta, \Sigma^i_\phi)\}_{i=1}^N$$

Decision Transformer



Reinforcement Learning via Sequence Modeling

Decision Transformer

- Reinforcement Learning via Sequence Modeling, where the input is

$$\tau = (\hat{R}_1, s_1, a_1, \hat{R}_2, s_2, a_2, \cdots, \hat{R}_T, s_T, a_T)$$

$$\{\hat{R}_t, S_t, a_t\}_{t=0}^{T} \qquad \hat{R}_t = \sum_{t'=t}^{T} r_{t'}$$

- Via autoregression, the generated output is

$$\{a_t\}_{t=0}^{T}$$

- The architecture of network is decoder only, masked multi-head self-attention.
- Position embedding: one timestep corresponds to three tokens (r,s,a)
- Embedding = embedding + position embedding

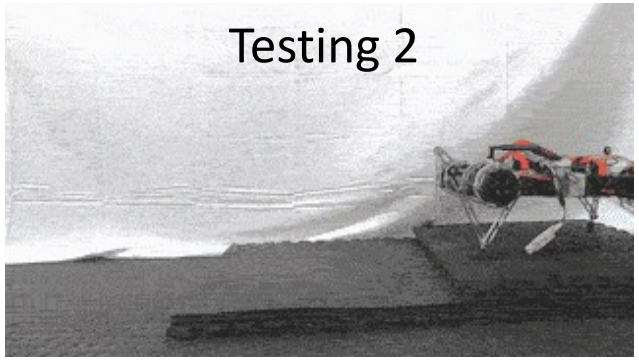
Outline

- ❑ Deep Reinforcement Learning
- ❑ **Applications to Robotics**
- ❑ Applications to LLMs

Applications to Robotics

- SAC for Robot Walking
- Policy Learning for Footed Robot
- Robot Manipulation

Soft Actor-Critic



Algorithm 1 Soft Actor-Critic

Initialize parameter vectors ψ, $\bar{\psi}$, θ, ϕ.
for each iteration **do**
  for each environment step **do**
    $\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$
    $\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_V \hat{\nabla}_\psi J_V(\psi)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$
    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$
    $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
  end for
end for

Update value V

Update Q

Update Policy

Update target value network

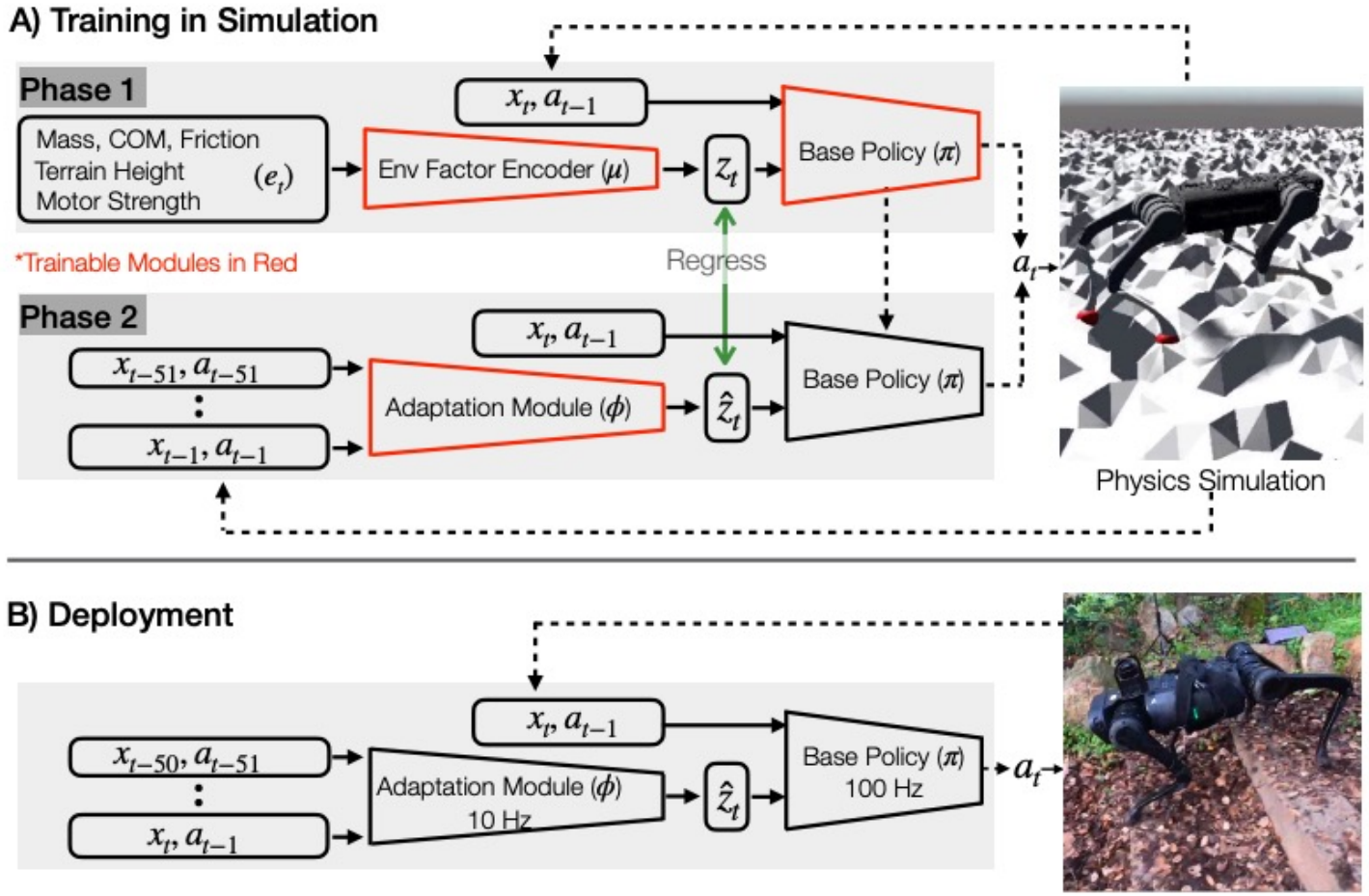
use the minimum of Q-functions for the value gradient

Domain Adaptation for Quadruped Robot

- **Unobservable Privileged Information**
  - a base policy
  - an adaptation module

- Trained on a varied terrain (simulated) generator using bioenergetics-inspired rewards.

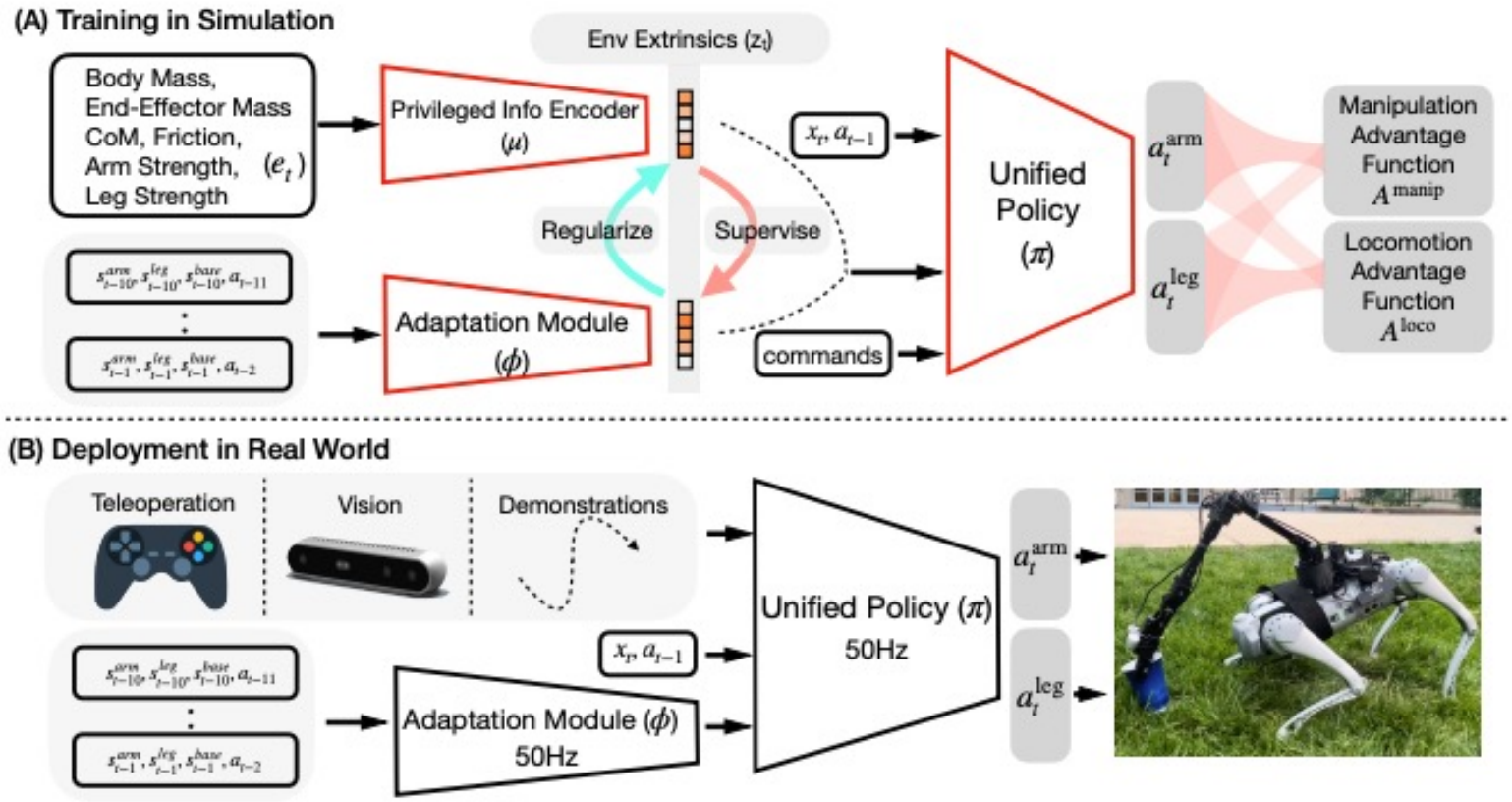
- Deployed on a variety of difficult terrains.



Domain Adaptation for Quadruped Robot

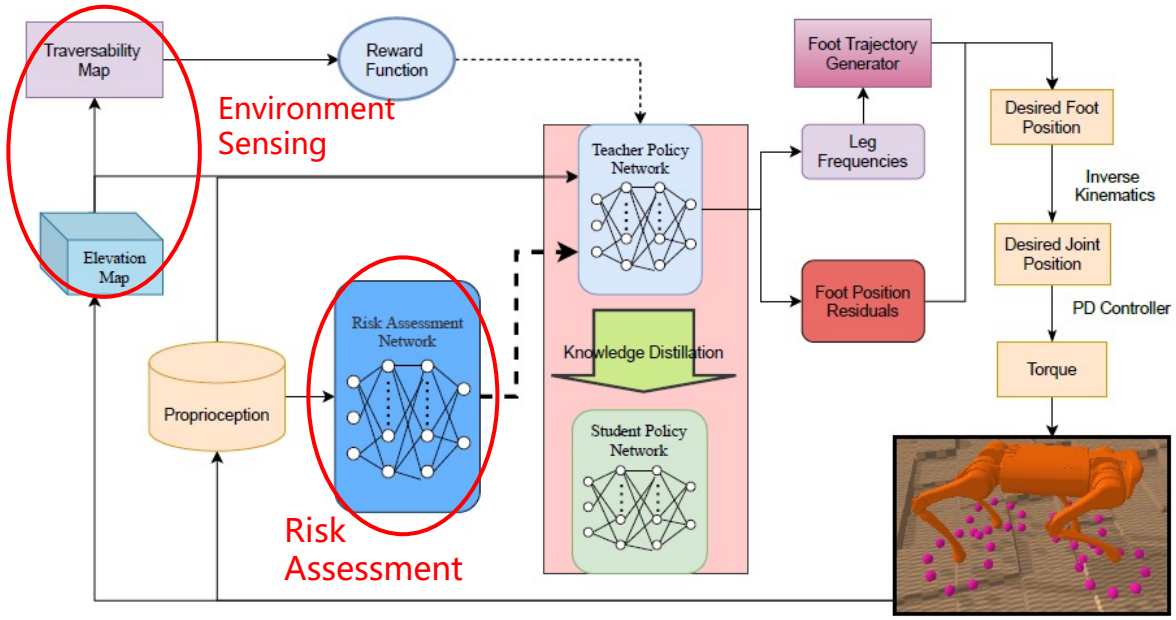
- **Unobservable Privileged Information**
  - a base policy
  - an adaptation module

- Mobile Manipulation, Whole-Body Control, Legged Locomotion



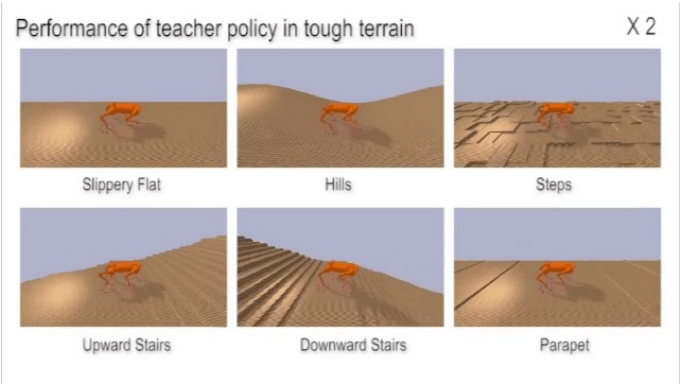
DRL-based Decision Strategy

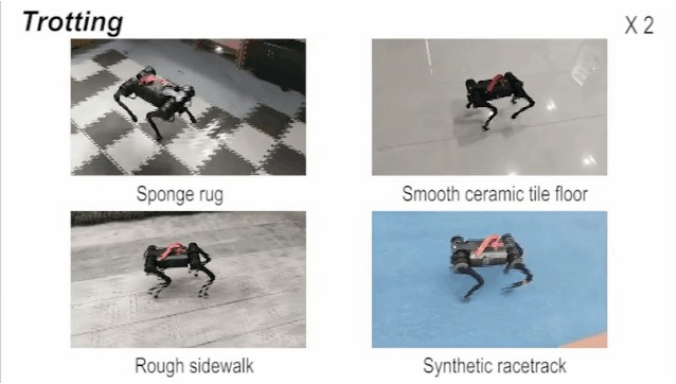
Risk Assessment Network(RAN) in DRL for safety locomotion



The RAN is incorporated into the model-free RL (e.g. SAC algorithm) as a penalty item δ to the loss function of the value and policy function.

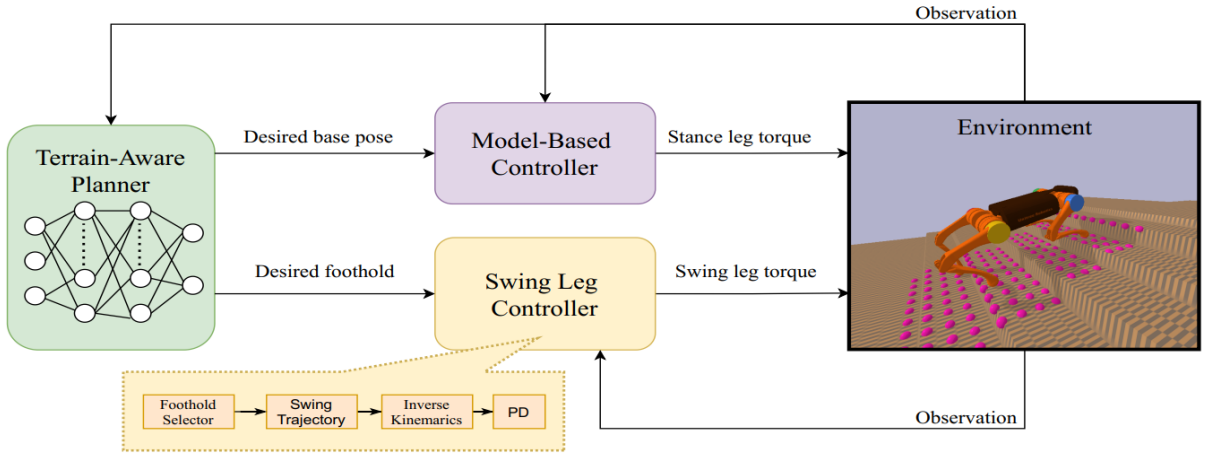
Result

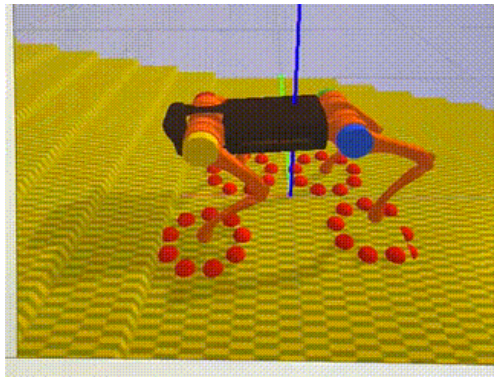


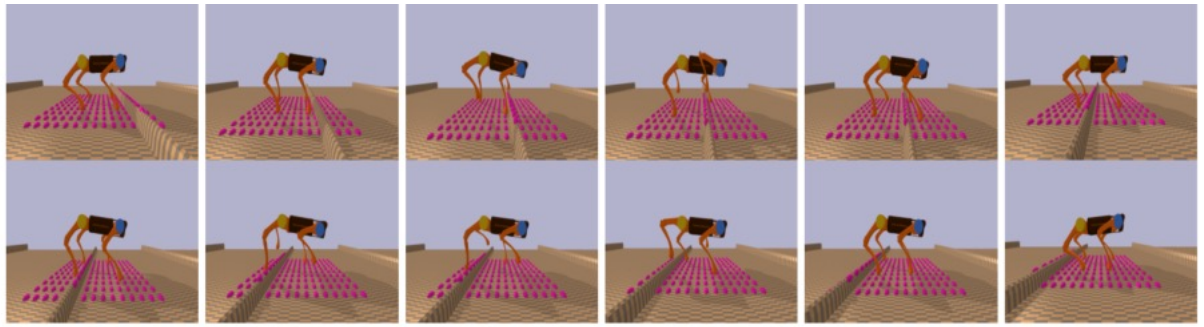


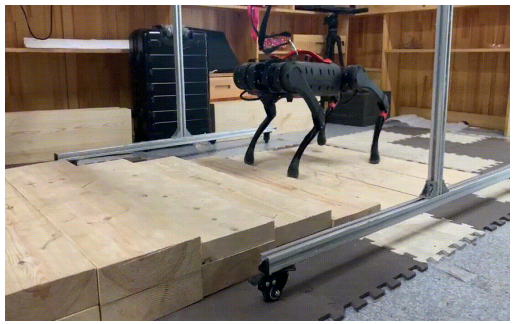
Hierarchical RL for Quadruped Robot

Hierarchical Reinforcement Learning Control Strategy



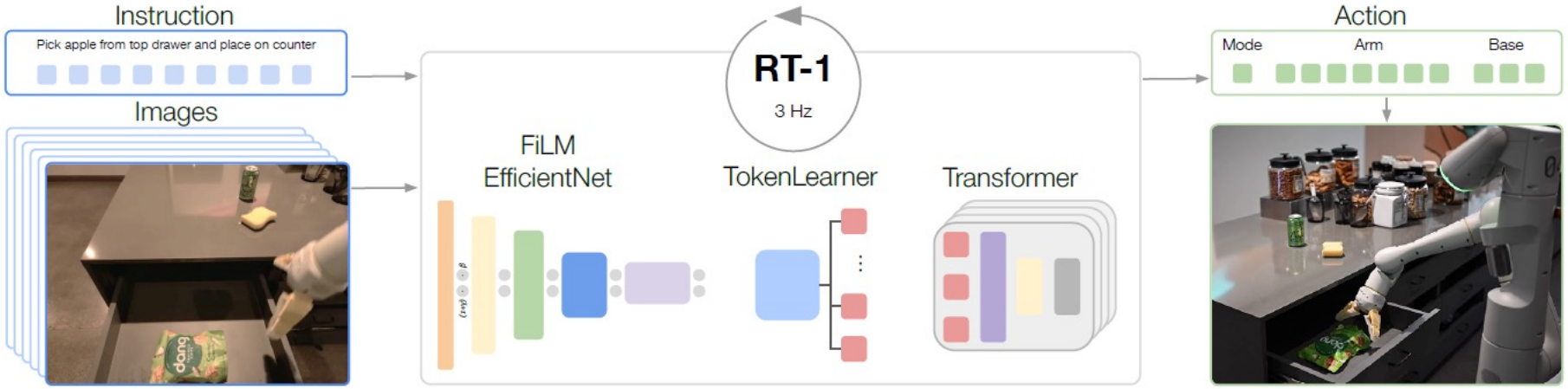






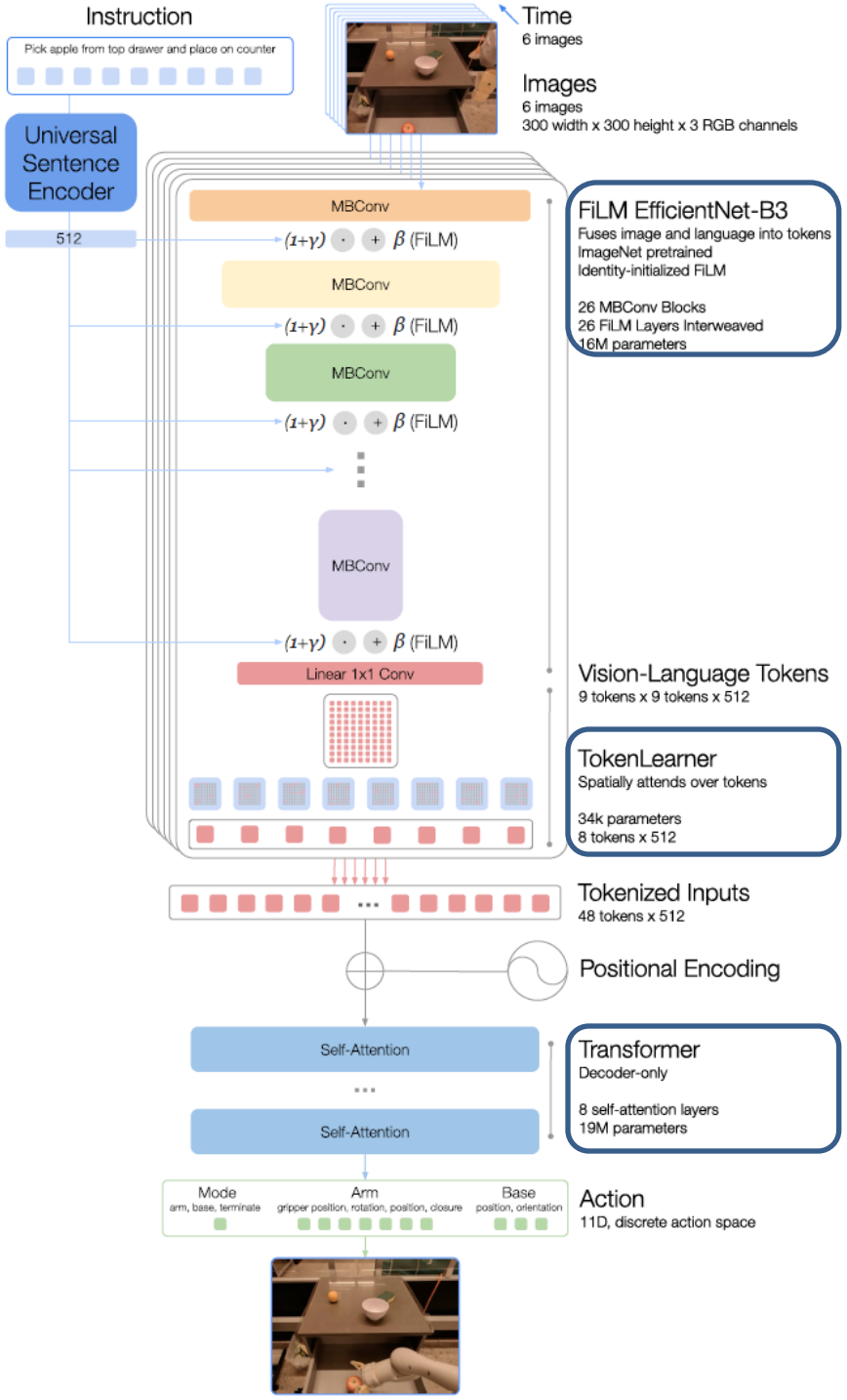
Quadruped **adjust the posture adaptively** varying the terrain changes

Robotics Transformer (RT-1)

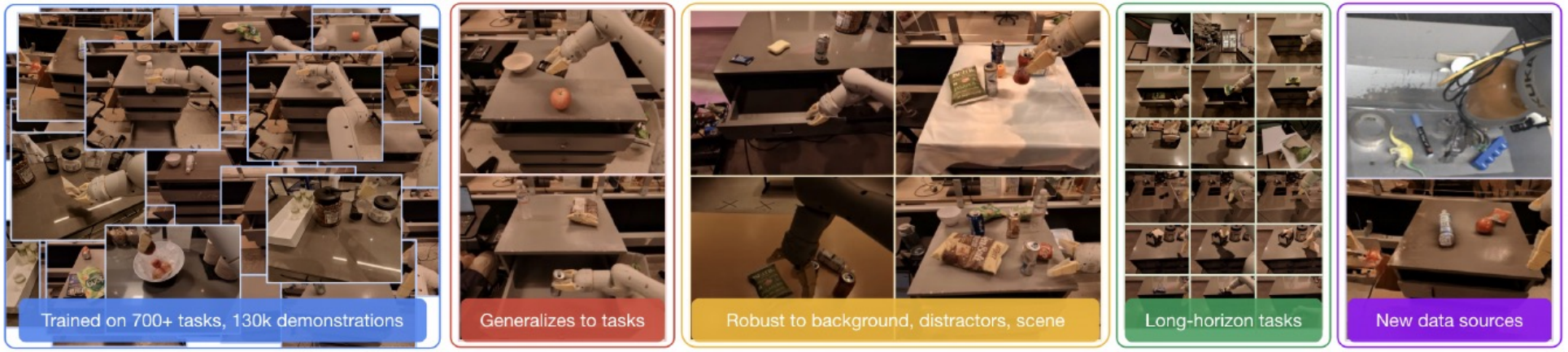


- RT-1 takes images and natural language instructions and outputs discretized base and arm actions.
- Despite its size (35M parameters), it does this at 3 Hz.
- Efficient yet high-capacity architecture:
  - A FiLM (Perez et al., 2018) conditioned EfficientNet (Tan & Le, 2019)
  - A TokenLearner (Ryoo et al., 2021)
  - A Transformer (Vaswani et al., 2017).

Robotics Transformer (RT-1)

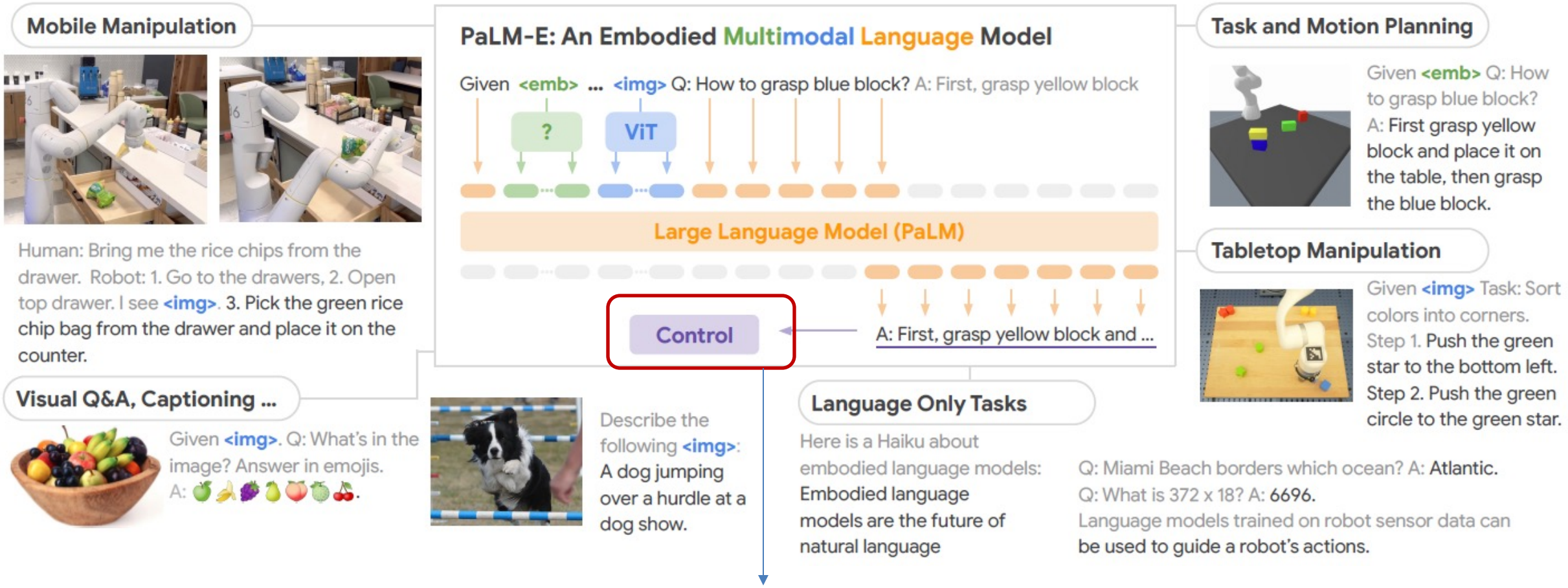


Robotics Transformer (RT-1)



- RT-1's large-scale, real-world training (130k demonstrations) and evaluation (3000 real-world trials)
- Impressive generalization, robustness, and ability to learn from diverse data

PaLM-E



Robotics Transformer (RT-1)

RT-2

- LLM + RL: RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control

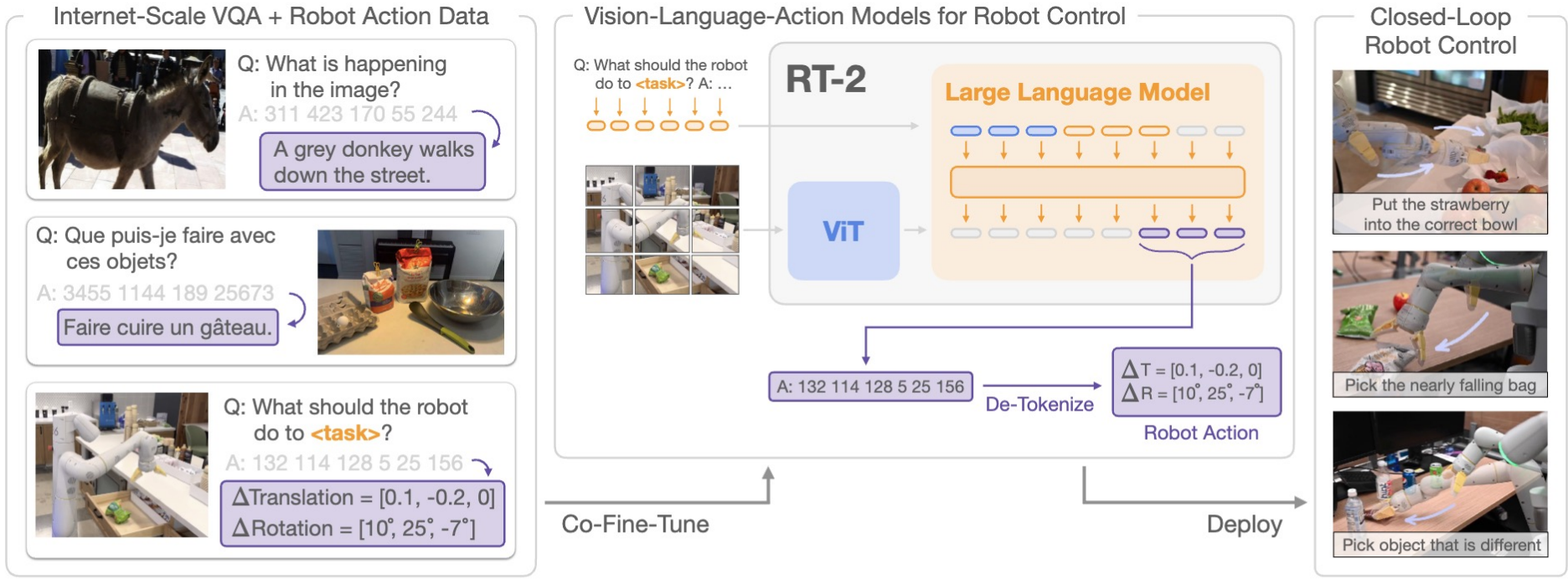
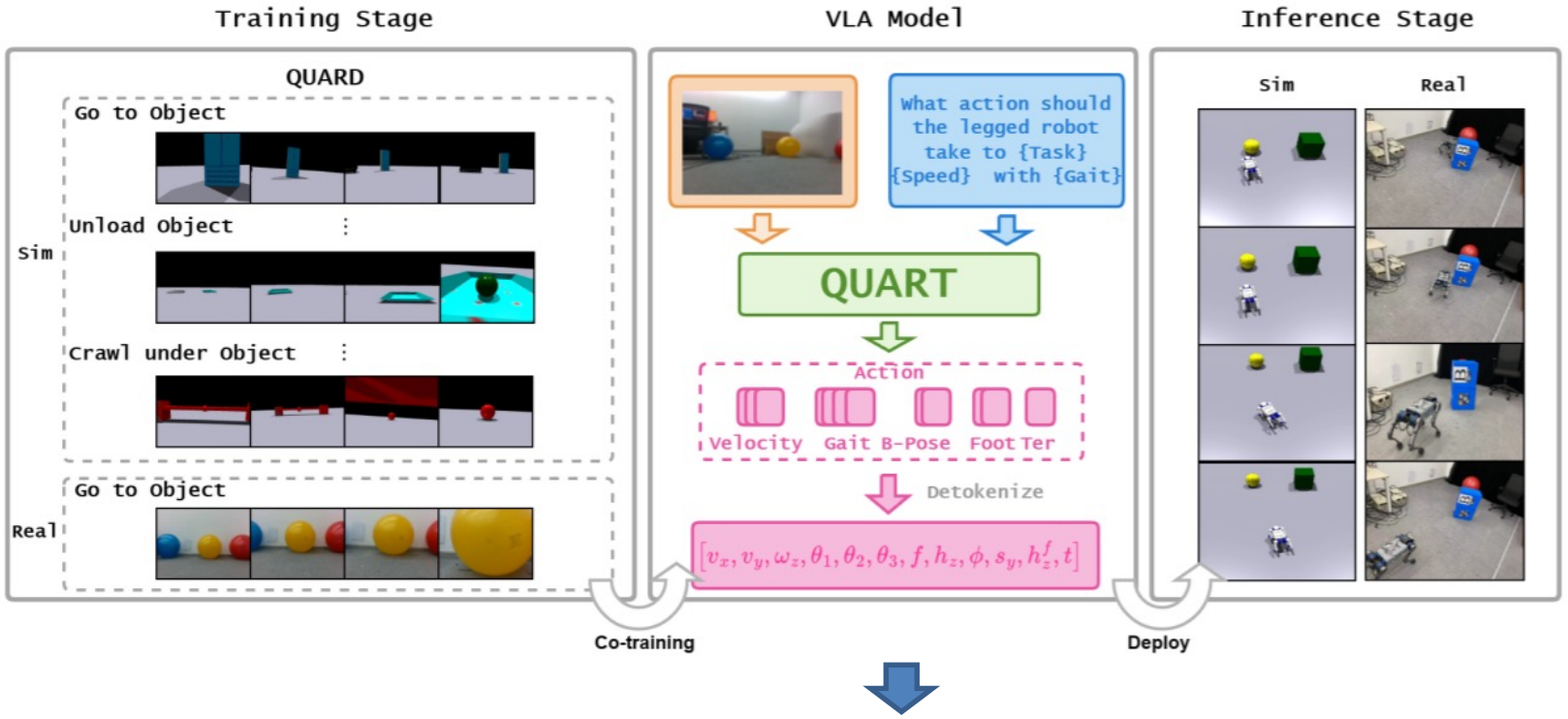


Figure 1 | RT-2 overview: we represent robot actions as another language, which can be cast into text tokens and trained together with Internet-scale vision-language datasets. During inference, the text tokens are de-tokenized into robot actions, enabling closed loop control. This allows us to leverage the backbone and pretraining of vision-language models in learning robotic policies, transferring some of their generalization, semantic understanding, and reasoning to robotic control. We demonstrate examples of RT-2 execution on the project website: robotics-transformer2.github.io.

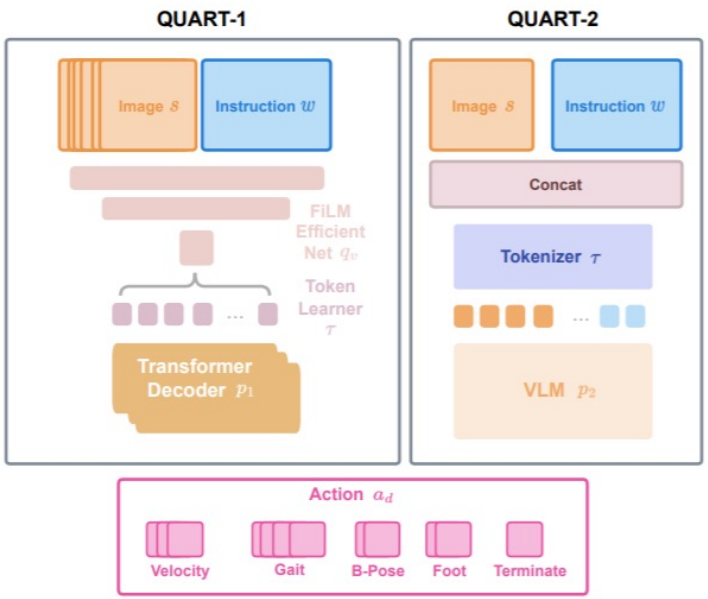

hint

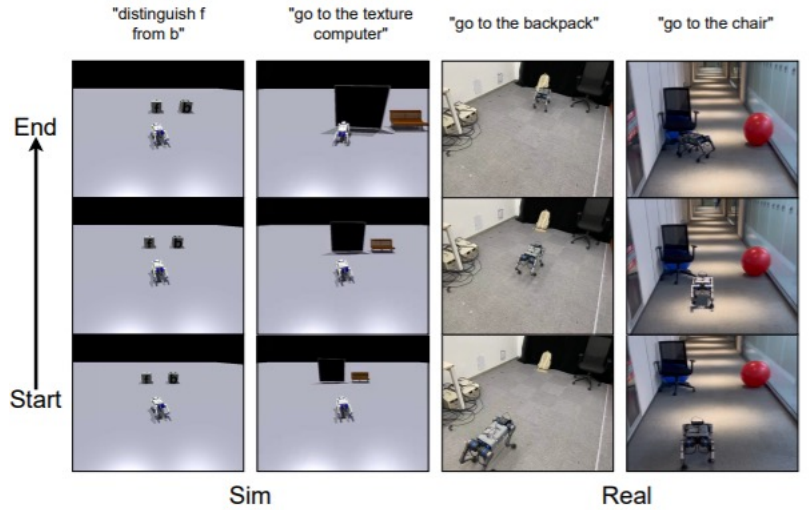
QUART

Architecture of QUART-1 and QUART-2



- By combining FiLM, TokenLearner, and a transformer decoder, QUART-1 generates discretized action tokens.
- In contrast, QUART-2 is designed to leverage the scene comprehension capability of a pretrained VLM.



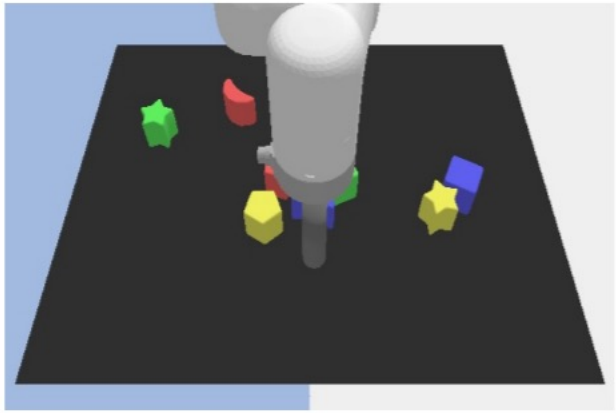


Cobra

LLaVA v1.5

MobileVLM v2

Cobra



Can you describe the image in detail?

*The image features a **black and white checkered floor** with a variety of colorful toys scattered across it. ..... Additionally, there are two **toy cars**, one located towards the left side of the floor and the other towards the right side. .....*

*The image presents a 3D rendering of a **game board**, which is the central focus of the image. ..... In the top right corner of the board, there's a **white ball**, which appears to be the central object of the game. .....*

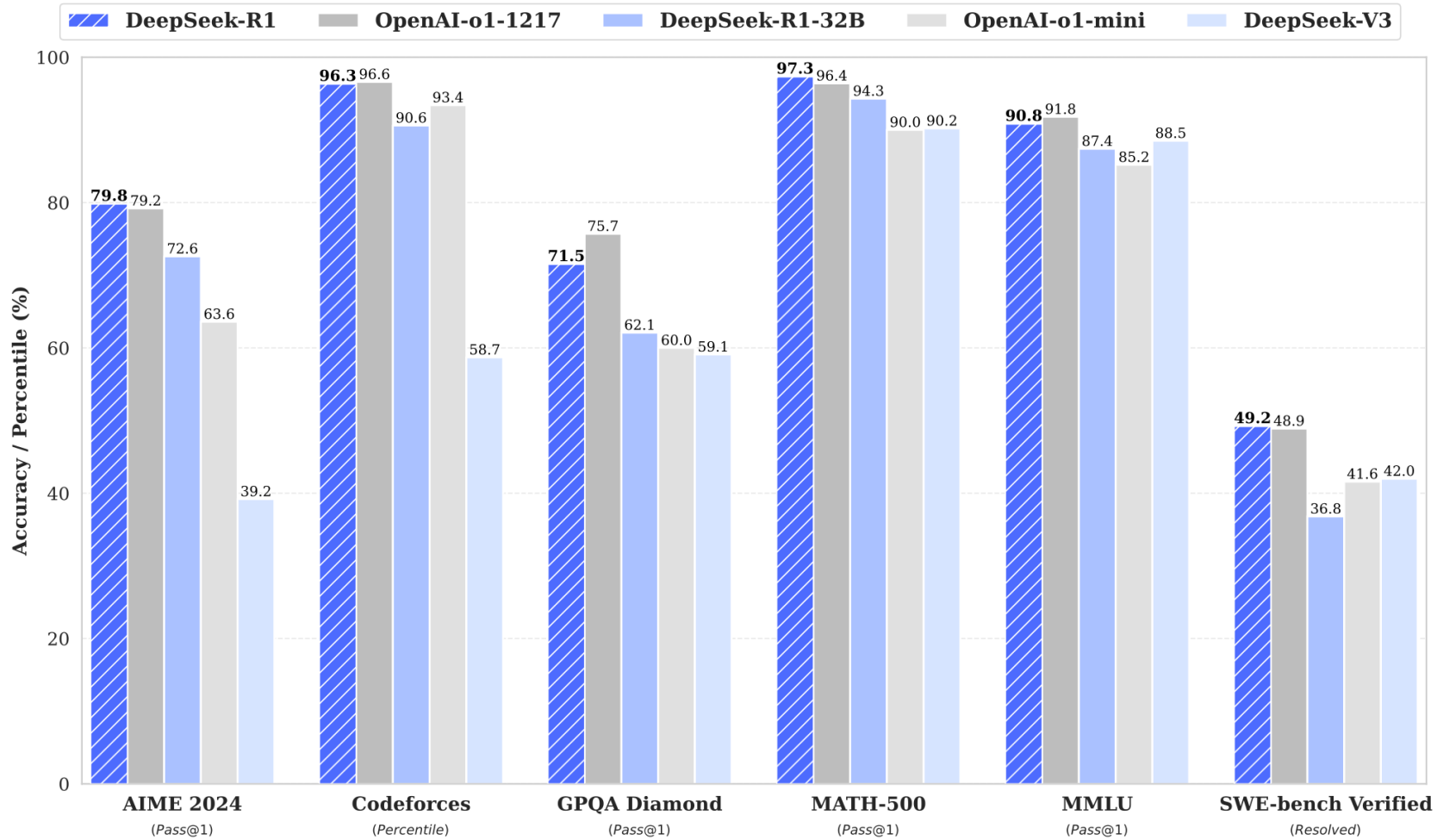
*The image shows a computer-generated scene with a white, cylindrical object, possibly a piece of machinery or a **robot**, surrounded by various colored blocks. The blocks are of different shapes and colors, including red, yellow, and blue. The scene appears to be a **computer-generated** image, possibly a 3D model or a digital artwork.The image is a close-up view of the object, emphasizing its cylindrical shape and the surrounding blocks.*

Outline

- ❑ Deep Reinforcement Learning
- ❑ Applications to Robotics
- ❑ Applications to LLMs

WESTLAKE UNIVERSITY

Remarkable performance of DeepSeek-R1:



Background: Improving LLM's reasoning ability

- OpenAI's o1 [1]: The first to introduce inference-time scaling by increasing the length of the Chain-of-Thought reasoning process.
  - It achieved significant improvements in various reasoning tasks, such as mathematics, coding, and scientific reasoning
- However, process-based reward models [2][3][4], reinforcement learning [5], and search algorithms such as Monte Carlo Tree Search [6] have not achieved comparable performance.

[1] OpenAI. Learning to reason with llms, 2024b. URL https://openai.com/index/learning-to-reason-with-llms/
[2] H. Lightman, et al. Let's verify step by step. arXiv preprint arXiv:2305.20050, 2023.
[3] J. Uesato, et al. Solving math word problems with process-and outcome-based feedback. arXiv preprint arXiv:2211.14275, 2022.
[4] P. Wang, et al. Math-shepherd: A label- free step-by-step verifier for llms in mathematical reasoning. arXiv preprint arXiv:2312.08935, 2023.
[5] A. Kumar, et al. Training language models to self-correct via reinforcement learning. arXiv preprint arXiv:2409.12917, 2024.
[6] X. Feng,. Alphazero-like tree-search can guide large language model decoding and training, 2024. URL https://arxiv.org/abs/2309.17179

Contribution of DeepSeek-R1

- The first open research to validate that reasoning capabilities of LLMs can be incentivized purely through RL, without the need for supervised fine-tuning (SFT).
  - **DeepSeek-R1-Zero:** applies RL directly to the base model without SFT. Achieved good performance but with poor readability and language mixing
  - **DeepSeek-R1:** applies RL from a checkpoint fine-tuned with thousands of long Chain-of-Thought (CoT) examples, achieves a score of 79.8% Pass@1 on AIME 2024, slightly surpassing OpenAI-o1-1217.
- Demonstrate that distilled smaller dense models perform exceptionally well
  - E.g., DeepSeek-R1-Distill-Qwen-7B achieves 55.5% on AIME 2024, surpassing QwQ-32B-Preview.

DeepSeek-R1-Zero: RL on the Base Model

Use **Group Relative Policy Optimization (GRPO) [1]:**

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}\left[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)\right]$$

$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right)$$

Comparing with **PPO:**

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right]$$

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, M. Zhang, Y. Li, Y. Wu, and D. Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. arXiv preprint arXiv:2402.03300, 2024.

Understanding GRPO

GRPO:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right)$$

q: each question

$o_1, o_2, \ldots o_G$: a group of outputs sampled from $\pi_{\theta_{old}}(o_i|q)$

Advantage:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}$$

computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs within each group

Understanding GRPO: KL term

Here they use $\mathbb{D}_{KL}(q||p) = \mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)} + \left(\frac{p(x)}{q(x)} - 1\right)\right] \approx \sum_{i=1}^{N}\left(-\log\frac{p(x_i)}{q(x_i)} + \left(\frac{p(x_i)}{q(x_i)} - 1\right)\right)$

Different from the standard $\mathbb{D}_{KL}(q||p) = \mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)}\right] \approx \sum_{i=1}^{N}\left(-\log\frac{p(x_i)}{q(x_i)}\right)$ Monte Carlo estimation $x_i \sim q(x)$

What makes a good estimator?

- Unbiased (has the right mean)
- Low variance

$\mathbb{D}_{KL}(q||p) = \mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)}\right]$ is unbiased, but has high variance (~half of $-\log\frac{p(x_i)}{q(x_i)}$ is negative)

How to reduce variance? Actor-critic: $\nabla_\theta J(\theta) = \mathbb{E}_{S\sim\eta, A\sim\pi(S;\theta)}\left[\nabla_\theta \ln\pi(A|S;\theta)\left(q_\pi(S,A) - v_\pi(S)\right)\right]$

KL term with variance reduction

The general way to lower variance is with a control variate, *i.e.*, add something that has expectation zero but is negatively correlated with the original variable.

The only interesting thing that's guaranteed to have zero expectation is $\mathbb{E}_{x\sim q(x)}\left[\frac{p(x)}{q(x)}-1\right]$, so we add it:

$$\mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)}+\lambda\left(\frac{p(x)}{q(x)}-1\right)\right]$$

How to choose λ?

One way is: we can do a calculation to minimize the variance of this estimator and solve for λ. But we get an expression that depends on p and q and is hard to calculate analytically.

Another way: Note that $\log x$ is concave, and $\log x \le x-1$. Therefore, when $\lambda=1$, $-\log\frac{p(x)}{q(x)}+\left(\frac{p(x)}{q(x)}-1\right)$ is always positive.

> *The idea of measuring distance by looking at the difference between a convex function and its tangent plane appears in many places. It's called a Bregman Divergence.*

KL term with variance reduction

$$\mathbb{D}_{KL}(q||p) = \mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)}\right] \longrightarrow \mathbb{E}_{x\sim q(x)}\left[-\log\frac{p(x)}{q(x)} + \lambda\left(\frac{p(x)}{q(x)} - 1\right)\right]$$

$$\mathbb{D}_{KL}(p||q) = \mathbb{E}_{x\sim q(x)}\left[\frac{p(x)}{q(x)}\log\frac{p(x)}{q(x)}\right] \longrightarrow \mathbb{E}_{x\sim q(x)}\left[\frac{p(x)}{q(x)}\log\frac{p(x)}{q(x)} - \left(\frac{p(x)}{q(x)} - 1\right)\right]$$

We can also do the variance reduction for general f-divergences:

$$\mathbb{D}_f(p||q) = \mathbb{E}_{x\sim q(x)}\left[f\left(\frac{p(x)}{q(x)}\right)\right] \longrightarrow \mathbb{E}_{x\sim q(x)}\left[f\left(\frac{p(x)}{q(x)}\right) - f'(1)\left(\frac{p(x)}{q(x)} - 1\right)\right]$$

$f: [0, +\infty) \to (-\infty, +\infty)$ is convex such that $f(1) = 0$ and $f(0) = \lim_{t\to 0^+} f(t)$

More at http://joschu.net/blog/kl-approx.html

DeepSeek-R1-Zero: Reward modeling

WESTLAKE UNIVERSITY

Accuracy reward:

Evaluates whether the response is correct.

- Math problems with deterministic results: Require to provide the final answer in a specified format
- LeetCode problems, a compiler can be used to generate feedback based on predefined test cases

They do not apply the outcome or process neural reward model due to *reward hacking*.

Format reward:

Enforces the model to put its thinking process between ‘<think>’ and ‘</think>’ tags

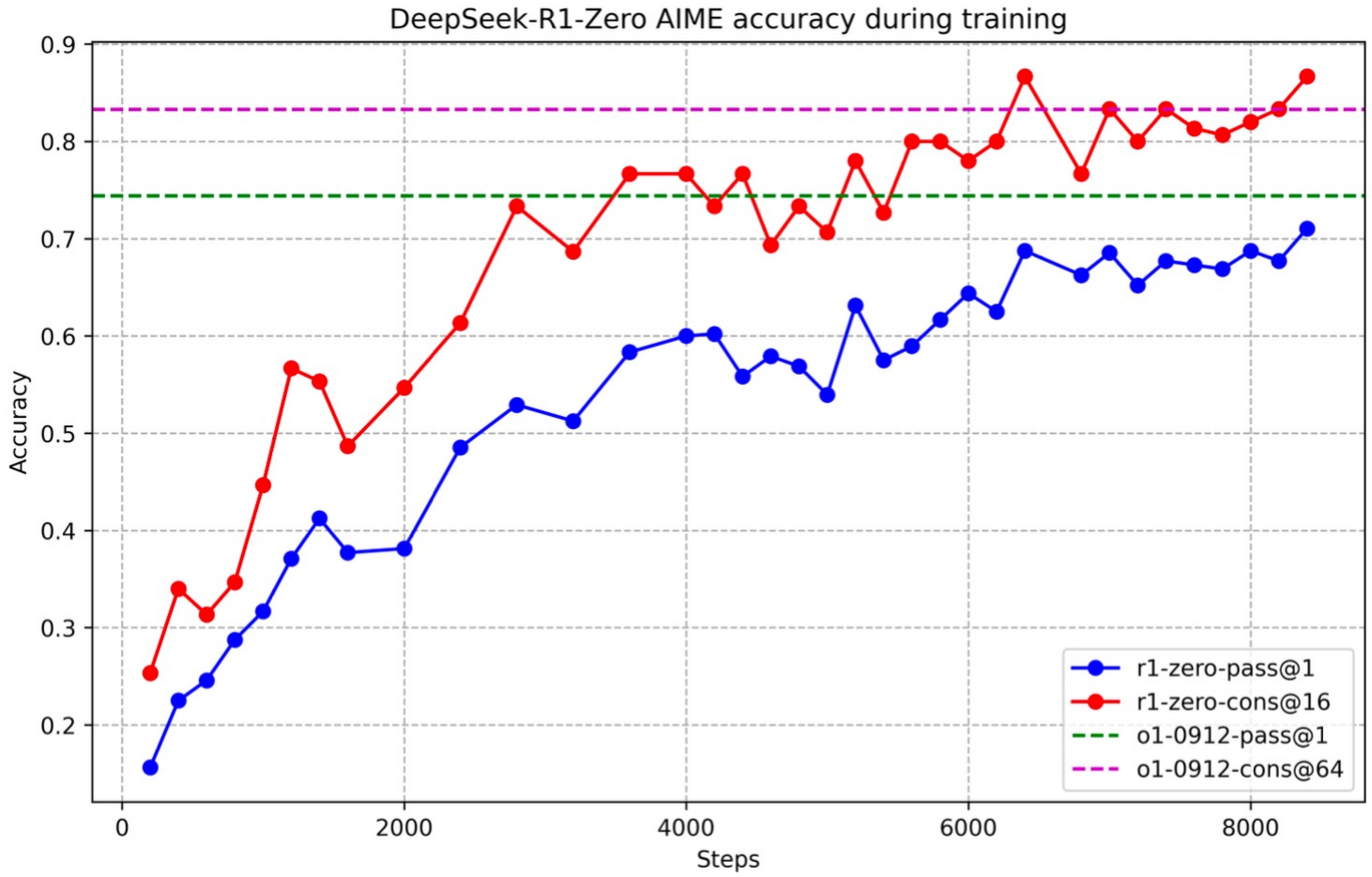
DeepSeek-R1-Zero: Training template

Training template:

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here </think> <answer> answer here </answer>. User: prompt. Assistant:

DeepSeek-R1-Zero: Performance

AIME accuracy of DeepSeek-R1-Zero during training:



DeepSeek-R1-Zero: Performance

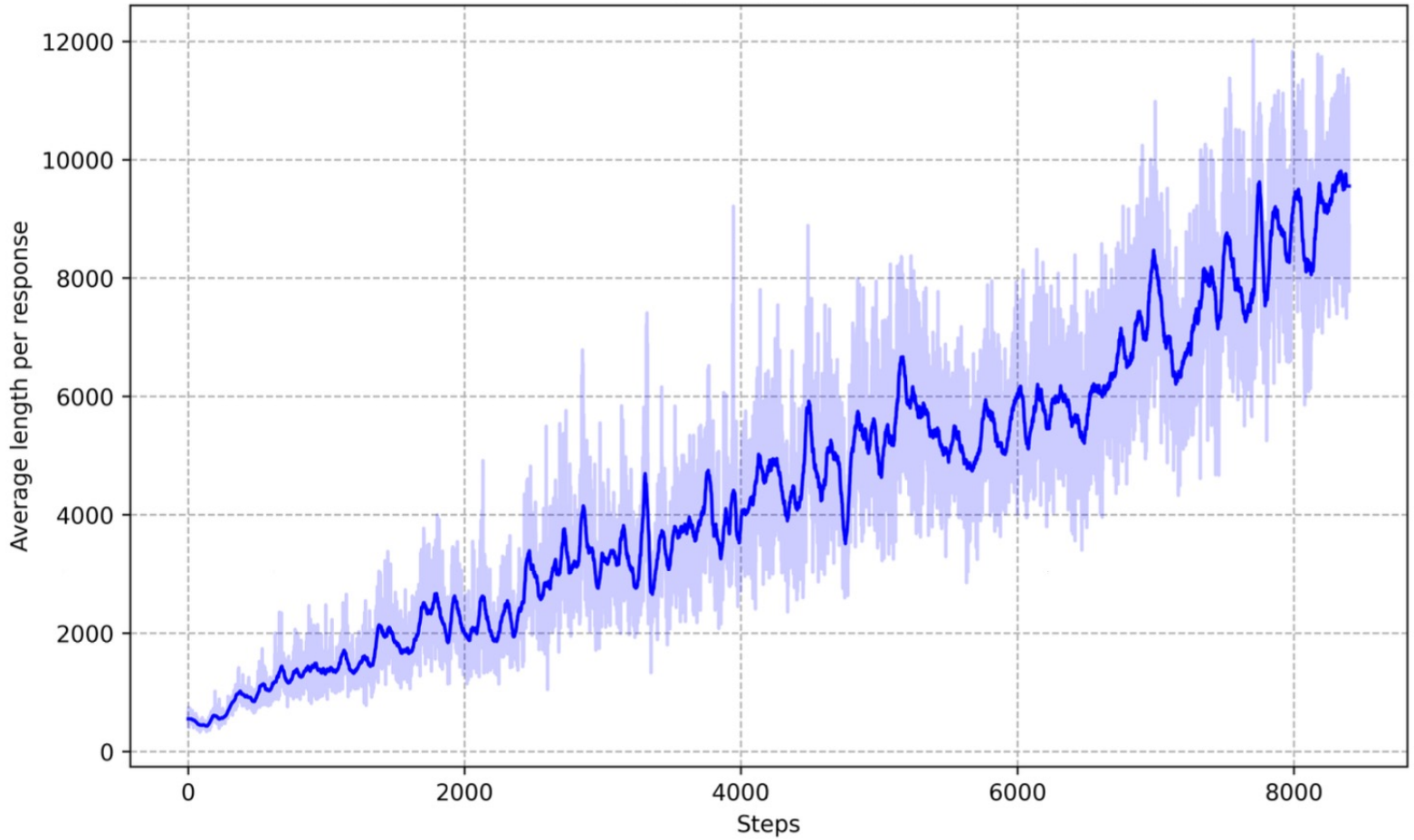
Comparison of DeepSeek-R1-Zero and OpenAI o1 models on reasoning-related benchmarks:

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

Show that DeepSeek-R1-Zero can attain robust reasoning capabilities **without the need for any supervised fine-tuning data.**

DeepSeek-R1-Zero: Average response length

The average response length of DeepSeek-R1-Zero on the training set during the RL process:



DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time

Emergence of sophisticated behaviors

Emergence of sophisticated behaviors:

- Reflection: The model revisits and reevaluates its previous steps
- The exploration of alternative approaches to problem-solving
- Aha moment

DeepSeek-R1-Zero: Aha moment

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\cdots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

$\cdots$

DeepSeek-R1

Drawback of DeepSeek-R1-Zero: Poor readability and language mixing

Solution:

- **Cold start**
  - Collect thousands of long CoT data to fine-tune the model as the initial RL actor. This improves readability and potential.
- **Reasoning-oriented RL**
  - Have a language consistency reward as the proportion of target language words in the CoT.
- **Rejection Sampling and Supervised Fine-Tuning**
  - Generate the data and fine-tune the model with reasoning trajectories by performing rejection sampling from the checkpoint from the above RL training, filtering out chain-of-thought with mixed languages, long paragraphs, and code blocks.
- **Reinforcement Learning for all Scenarios**
  - Secondary reinforcement learning stage aimed at improving the model's helpfulness and harmlessness while simultaneously refining its reasoning capabilities.

DeepSeek-R1: Performance

	Benchmark (Metric)	Claude-3.5-Sonnet-1022	GPT-4o 0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# Total Params	-	-	671B	-	-	671B
English	MMLU (Pass@1)	88.3	87.2	88.5	85.2	**91.8**	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	**92.9**
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	**84.0**
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	**92.2**
	IF-Eval (Prompt Strict)	**86.5**	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	**75.7**	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	**47.0**	30.1
	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	**82.5**
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	**87.6**
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	**92.3**
Code	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	**65.9**
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	**96.6**	96.3
	Codeforces (Rating)	717	759	1134	1820	**2061**	2029
	SWE Verified (Resolved)	**50.8**	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	**61.7**	53.3
Math	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	**79.8**
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	**97.3**
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	**78.8**
Chinese	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	**92.8**
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	**91.8**
	C-SimpleQA (Correct)	55.4	58.7	**68.0**	40.3	-	63.7

DeepSeek-R1: Other findings

Have tried some other approaches and has not worked yet:

- Process Reward Model (PRM)
- Monte Carlo Tree Search (MCTS)

Useful readings

Main paper: Guo, Daya, et al. "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning." *arXiv preprint arXiv:2501.12948* (2025).

Extension reading: DeepSeek r1闭门学习讨论_ 拾象

Related papers of RL for LLM:

- Training language models to follow instructions with human feedback 2022 （RLHF,PPO）
- RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback 2023
- Direct Preference Optimization: Your Language Model is Secretly a Reward Model 2023 (DPO)
- Reflexion: Language Agents with Verbal Reinforcement Learning 2023
- GPT-4 Technical Report 2023
- Offline Regularised Reinforcement Learning for Large Language Models Alignment 2024
- ORPO: Monolithic Preference Optimization without Reference Model 2024
- Reinforcement Learning for Aligning Large Language Models Agents with Interactive Environments: Quantifying and Mitigating Prompt Overfitting 2024
- DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning 2025 (GRPO)
- Blog: Reinforcement Learning in Large Language Models: A Technical Overview

Conclusion

- DRL basics and Model-free DRL
- Model-based DRL
- Inverse Reinforcement Learning
- Offline Reinforcement Learning
- Large Pre-training DRL Model
- Applications to Robotics: Robot Arm and Footed Robot
- Applications to LLM: DeepSeek-R1